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Master Thesis

2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach

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Declaration of Authorship

I, Leopold BODAMER, declare that this thesis titled, “2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while applying for a research degree at this University.
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- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
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Contents

Notation and conventions

Throughout this thesis, natural units are used such that $\hbar = 1$ and $c = 1$. Frequencies, energies, and inverse times are therefore expressed in the same units.

The following notation conventions apply:

- **Vectors.** Vectors in real space are denoted by bold symbols, e.g. $\boldsymbol{E}$ (electric field), $\boldsymbol{P}$ (polarisation), $\boldsymbol{k}$ (wave vector), $\boldsymbol{d}$ (transition dipole moment), and $\boldsymbol{e}$ (polarisation unit vector). The magnitude of a vector $\boldsymbol{d}$ is written as $d = |\boldsymbol{d}|$ when required.

- **Operators and states.** Operators acting on Hilbert spaces carry a hat, e.g. $\hat{H}$ for the Hamiltonian and $\hat{\boldsymbol{\mu}}$ for the dipole operator. Density operators are written as $\hat{\rho}$. Matrix elements in a chosen basis are denoted by plain symbols, e.g. $\rho_{ij} = \langle i|\hat{\rho}|j\rangle$.

- **Quantum mechanical pictures.** Different dynamical representations (Schrödinger, interaction, Heisenberg) are indicated by subscripts. For example, $\hat{O}_I(t)$ denotes an operator in the interaction picture. To avoid ambiguity with this convention, the interaction Hamiltonian is written as $\hat{H}_{\text{int}}$.

- **Superoperators.** Linear maps acting on operators (e.g. propagators or Liouvillians) are written in calligraphic font, such as $\mathcal{U}$, $\mathcal{L}$. Their action on density operators is written explicitly as $\mathcal{L}[\hat{\rho}]$.

- **Field projections.** For a fixed laboratory polarisation direction $\boldsymbol{e}$, vector fields are often projected onto this axis. The scalar field envelope is then defined as

  $$\varepsilon(t) = \boldsymbol{e} \cdot \boldsymbol{E}(t),$$

  and correspondingly $\hat{\mu} = \hat{\boldsymbol{\mu}} \cdot \boldsymbol{e}$. Whenever such projections are used, the underlying vector character is understood but suppressed in the notation.

- **Complex quantities.** The symbols Re and Im denote the real and imaginary parts of a complex quantity, respectively.

EBERHARD KARLS UNIVERSITÄT TÜBINGEN

Abstract

Mathematisch-Naturwissenschaftliche Fakultät
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2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach

by Leopold Bodamer

This thesis investigates the application of two-dimensional photon-echo spectroscopy to model quantum systems, with a focus on open quantum system dynamics. A comprehensive theoretical framework is developed that combines the response-function formalism of nonlinear spectroscopy in a non-perturbative setting with master-equation approaches for open quantum systems (QuTiP). Starting from minimal reference models, including a single qubit and a four-level system formed by two coupled qubits, the methods are validated and the approach is applied to a system of N qubits arranged on a cylindrical geometry inspired by the structural motifs of microtubules. The results demonstrate the capability of two-dimensional spectroscopy to probe coherence and energy-transfer dynamics in these systems, providing insights into their quantum behavior and interactions with the environment. This work lays the groundwork for future experimental and theoretical studies in the field of quantum biology and quantum information science.

Chapter 1

Introduction

REFERENCE: sounds good but are refs good?

Quantum dynamics in molecular and condensed-matter systems unfold on femtosecond timescales and are shaped by the interplay of coherent inter-site couplings and environmental fluctuations. In many systems of current interest—light-harvesting complexes, molecular aggregates, conjugated polymers—the relevant electronic degrees of freedom are embedded in warm, structurally complex surroundings that drive rapid decoherence and dissipation [1, 2].

Understanding how these environment-induced processes compete with coherent energy transfer, and identifying their signatures in experimentally accessible observables, is a central challenge in the physics of open quantum systems.

Conventional linear spectroscopy—absorption and emission—reveals excitation energies and oscillator strengths but compresses all dynamical information into one-dimensional lineshapes, where homogeneous dephasing, population relaxation, and static disorder produce overlapping contributions that are difficult to disentangle [3].

Nonlinear spectroscopy overcomes this limitation. In particular, two-dimensional electronic spectroscopy (2DES) correlates excitation and detection frequencies by controlling the time delays between a sequence of ultrashort laser pulses [4–6]. The resulting frequency–frequency maps separate homogeneous from inhomogeneous broadening, resolve inter-state couplings through cross peaks, and track energy-transfer pathways as a function of the waiting time. On the theory side, the measured signal can be expressed through multi-time response functions of the transition-dipole operator, providing a direct link between spectroscopic observables and open-quantum-system dynamics [3].

This thesis develops a computational framework for simulating 2DES signals of open quantum systems in parameter regimes relevant to warm, structured environments, and applies it to model systems of increasing complexity.

1.1 Scope and approach

The open-system dynamics are modelled with the Redfield master equation [1, 2, 7], which provides a transparent microscopic link from a system–bath Hamiltonian—characterised by bath correlation functions and spectral densities—to relaxation and dephasing rates. Because Redfield theory is not guaranteed to preserve complete positivity, its application is restricted to regimes where the weak-coupling and Markov assumptions are well motivated. Results are benchmarked against Lindblad-form generators to assess the impact of the secular approximation and to ensure physical consistency.

On the spectroscopy side, the third-order response-function formalism [3, 5] provides the conceptual framework. Numerically, however, a non-perturbative time-domain strategy is adopted: the driven open-system equation of motion is propagated with the laser pulses included explicitly, and lower-order contributions are eliminated through phase cycling over auxiliary propagations [8, 9]. This approach naturally accounts for finite pulse durations and pulse-overlap effects without resorting to the impulsive limit.

1.2 Model systems and physical questions

The formalism is applied to a hierarchy of model systems designed to isolate specific spectroscopic signatures:

- **Two-level system.** A minimal benchmark for validating the numerical framework and for disentangling the 2D lineshape contributions of homogeneous dephasing, population relaxation, and static disorder.
- **Dimer.** Two coupled sites that produce cross peaks, enabling a direct comparison of coherent excitonic mixing and incoherent population transfer during the waiting time.
- **N-site cylindrical ensemble.** A multi-chromophore system whose geometry is motivated by the protofilament arrangement in microtubules [10], serving as a structured test case for geometry-dependent coupling networks and their manifestation in 2D spectra.

The central physical question is how environment-induced relaxation and dephasing, as parametrised by the spectral density and temperature, compete with coherent couplings to determine peak positions, lineshapes, and waiting-time dynamics in two-dimensional spectra.

1.3 Thesis outline

Chapter 2 introduces the open-quantum-system framework, deriving the Redfield master equation and discussing bath correlation functions and spectral densities. Chapter 3 develops the nonlinear-spectroscopy formalism, from linear response and the Kubo formula through third-order response functions to the construction of rephasing and nonrephasing 2D spectra. Chapter 4 specifies the Hamiltonians, dipole operators, and bath models for the systems listed above.

Technical supporting material is collected in the appendices: App. A (numerical implementation and software architecture), App. C (Maxwell-to-signal derivation for coherent 2D Fourier-transform spectroscopy), and App. B (Dyson-series derivation of the n-th order polarisation).

Chapter 2

Open Quantum Systems

Any real-world quantum system is not perfectly isolated. Instead, it is coupled to surrounding degrees of freedom such as electromagnetic modes, lattice vibrations, or solvent fluctuations, though the coupling is often weak but never exactly zero. In contrast to an ideal closed system with unitary Schrödinger evolution,

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle = \hat{H}(t)\left|\psi(t)\right\rangle, \tag{2.1}$$

an open system generally exhibits effective non-unitary dynamics once the environment is traced out. This coupling drives relaxation toward an environment-defined equilibrium and leads to two key mechanisms: **decoherence**, the loss of phase relations and dephasing, and **dissipation**, energy exchange and thermalization [1, 2].

Concrete examples include atoms subject to vacuum electromagnetic field fluctuations [1], quantum dots in solid-state settings coupled to phonons [11], molecular systems interacting with surrounding solvent degrees of freedom [3], and quantum computers whose coherence can be rapidly degraded by environmental noise [12].

For molecular chemistry and biological complexes, the relevant quantum degrees of freedom are embedded in warm, wet, and structurally complex environments, so strong decoherence and fast relaxation are expected. Yet these systems must still obey quantum laws, so the task is to explain classical-looking behavior within a fully quantum description. [13].

This chapter provides the theoretical foundation for including environmental effects in the quantum-dynamical models used throughout the thesis. A practical weak-coupling description is introduced that connects microscopic Hamiltonians to relaxation/dephasing rates and to bath models used later in applications and simulations. In Sec. 2.1 the Redfield master equation is introduced and derived from microscopic system–bath dynamics. In Sec. 2.2 bath correlation functions and their Fourier/spectral representations are discussed, including how emission and absorption arise via detailed balance. In Sec. 2.3 harmonic-oscillator (bosonic) environments are considered and explicit thermal correlators are computed, establishing the link to spectral densities. Finally, Sec. 2.4 summarizes common phenomenological spectral densities such as Ohmic-type and Drude–Lorentz and their physical interpretation.

2.1 The Redfield Master Equation

In this thesis the Redfield master equation is used as the weak-coupling description of open-system dynamics. Once the environment is specified by its correlation function or equivalently a spectral density, the reduced dynamics follows directly from the microscopic system–bath model and can be simulated efficiently even for multi-level molecular and excitonic systems [1–3]. More accurate methods such as numerically exact path-integral [14] or hierarchy techniques [15] can be too costly for the parameter scans and system sizes considered here. It is important to note that the Redfield equation, while preserving trace and Hermiticity, does not guarantee complete positivity of the density matrix, which is a fundamental requirement for physical quantum states [7]. So care must be taken when determining when the Redfield equation is useful [16–18].

Originally developed by A.G. Redfield in 1957 and 1965 for nuclear magnetic resonance relaxation phenomena [16], the Redfield equation describes the time evolution of the reduced density matrix, formally defined in Sec. 2.1.1, of a quantum system weakly coupled to a thermal environment. Unlike phenomenological approaches that introduce dissipation and dephasing *ad hoc*, the Redfield equation relates them to environmental correlation functions or spectral densities [1, 2, 19].

The following requirements must be fulfilled by the final derived Redfield equation:

1. The equation should be linear in the system density matrix $\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S}}(t) = F(\hat{\rho}_{\mathrm{S}}(t))$. This is the reduced equation of motion.

2. The equation should be Markovian, meaning that the evolution of the system density matrix at time t only depends on the state of the system at time t and not on its past history.

3. The equation should be trace-preserving. $\mathrm{Tr}[\hat{\rho}_{\mathrm{S}}(t)] = \mathrm{Tr}[\hat{\rho}_{\mathrm{S}}(0)]$ for all times t.

The Redfield equation is now derived from the microscopic dynamics of the system–environment composite, following the approach presented in [20] and standard textbooks such as Breuer and Petruccione [1]. The steps are presented in a more explicit way. In particular, the elimination of the first-order contribution is treated rigorously. The environmental correlation functions and spectral densities that characterize the bath properties and determine the system's relaxation behavior are then examined.

2.1.1 Mathematical Background

Before the derivation of the Redfield equation, the basic mathematical tools and notation used throughout are briefly collected: the density-operator formalism for mixed states, the trace and partial trace for extracting reduced system dynamics, and the corresponding expectation values.

Density Matrix Formalism

The density matrix formalism provides a powerful framework to describe the dynamics of quantum systems, especially when dealing with mixed states and decoherence effects. The density matrix $\hat{\rho}$ is defined as [21]

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|, \quad p_i \geq 0, \quad \sum_i p_i = 1. \tag{2.2}$$

It represents a statistical mixture, the average of an ensemble with different realizations of pure quantum states ψ_i with classical probabilities p_i. In other words, one cannot tell with certainty the quantum state of the system. This is particularly useful in the context of open quantum systems, where the system is entangled with an environment and one loses information about the system of interest to the total system. The conditions in Eq. (2.2) ensure the following properties that a physical density matrix must satisfy

$$\begin{aligned} \hat{\rho}^\dagger &= \hat{\rho}, \\ \left\langle\psi\right| \hat{\rho} \left|\psi\right\rangle &\geq 0 \quad \forall \left|\psi\right\rangle, \\ \mathrm{Tr}(\hat{\rho}) &= 1. \end{aligned}$$

In this formalism, for a canonical basis $\{|i\rangle\}$, the diagonal elements ρ_{ii} represent populations and the off-diagonal elements ρ_{ij} represent coherences between states. Coherences are phase relations, which are crucial for interference. For example for a two level system a clear phase relation and a pure quantum state would have off diagonal elements of $\rho_{ij} = 1/2$. This state is often referred to as a fully coherent superposition of the two states.

When coupling a system to an environment, the environment is responsible for decoherence. The state evolves over time to a purely statistical mixture of states.

The mean value or expectation value of an operator $\hat{O}$ can be calculated by additionally averaging over the classical probabilities p_i to obtain

$$\begin{aligned}\langle\hat{A}\rangle &= \sum_i p_i \langle\psi_i|\,\hat{A}\,|\psi_i\rangle \\ &= \sum_n\sum_i p_i \langle\psi_i|\,\hat{A}\,|\phi_n\rangle\langle\phi_n|\psi_i\rangle \\ &= \sum_n \langle\phi_n|\sum_i |\psi_i\rangle\, p_i \langle\psi_i|\,\hat{A}\,|\phi_n\rangle \\ &= \mathrm{Tr}(\hat{\rho}\,\hat{A}).\end{aligned}$$

Here, an arbitrary basis $\{|\phi_n\rangle\}$ and a corresponding completeness relation $\sum_n |\phi_n\rangle\langle\phi_n| = \mathbb{1}$ was used.

Partial Trace

The partial trace simplifies a quantum system description by removing certain parts through averaging. It can be viewed as the inverse operation to forming a tensor-product space. This is useful when only one part of a composite system is of interest. For a bipartite Hilbert space $\mathcal{H}_\mathcal{A}\otimes\mathcal{H}_\mathcal{B}$, the partial trace over $\mathcal{H}_\mathcal{B}$ is the unique linear map $\mathrm{Tr}_B : \mathcal{B}_1(\mathcal{H}_\mathcal{A}\otimes\mathcal{H}_\mathcal{B}) \to \mathcal{B}_1(\mathcal{H}_\mathcal{A})$ satisfying $\mathrm{Tr}[(\mathrm{Tr}_B\hat{X})\,\hat{A}] = \mathrm{Tr}[\hat{X}(\hat{A}\otimes\mathbb{1}_B)]$ for all $\hat{A}\in\ \mathcal{B}(\mathcal{H}_\mathcal{A})$, where $\mathcal{B}(\mathcal{H})$ is the algebra of bounded operators on $\mathcal{H}$, and $\mathcal{B}_1$ denotes trace-class operators, namely those X with finite trace norm[1, 22]

$$\|\hat{X}\|_1 = \mathrm{Tr}\sqrt{\hat{X}^\dagger\hat{X}}$$

This ensures expectation values of local observables $\hat{A}\otimes\mathbb{1}_B$ are preserved. In an orthonormal basis $\{|b_i\rangle\}$ of $\mathcal{H}_\mathcal{B}$, it acts as

$$\mathrm{Tr}_B\hat{X} = \sum_i(\mathbb{1}_A\otimes\langle b_i|)\,\hat{X}\,(\mathbb{1}_A\otimes|b_i\rangle). \tag{2.3}$$

With this definition thermal expectation values take the form

$$\langle\hat{A}\rangle = \mathrm{Tr}[\hat{\rho}\,\hat{A}] = \frac{1}{Z}\sum_n e^{-\beta E_n}A_{nn}, \tag{2.4}$$

with the inverse temperature

$$\beta = \frac{1}{k_\mathrm{B}T}.$$

2.1.2 System–Environment Setup

A quantum system of interest interacting with an environment with effectively infinitely many degrees of freedom is considered. The total Hilbert space $\mathcal{H}_\mathrm{T}$ is the tensor product of the Hilbert space of the system $\mathcal{H}_\mathrm{S}$ and that of the environment $\mathcal{H}_\mathrm{E}$:

$$\mathcal{H}_\mathrm{T} = \mathcal{H}_\mathrm{S}\otimes\mathcal{H}_\mathrm{E}.$$

The evolution of the total system is governed by the Liouville–von Neumann equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_\mathrm{T}(t) = -\mathrm{i}[\hat{H}_\mathrm{T},\hat{\rho}_\mathrm{T}(t)], \tag{2.5}$$

where $\hat{\rho}_{\mathrm{T}}(t)$ is the density matrix of the total system and $\hat{H}_{\mathrm{T}}$ is the total Hamiltonian. Units with $\hbar = c = 1$ are used throughout this thesis. Equation (2.5) is the mixed-state analogue of the Schrödinger equation Eq. (2.1) (see Sec. 2.1.1). Without loss of generality, the total Hamiltonian $\hat{H}_{\mathrm{T}} \in \mathcal{B}(\mathcal{H}_{\mathrm{T}})$ can be decomposed as

$$\hat{H}_{\mathrm{T}} = \hat{H}_{\mathrm{S}} \otimes \mathbb{1}_{\mathrm{E}} + \mathbb{1}_{\mathrm{S}} \otimes \hat{H}_{\mathrm{E}} + \alpha \hat{H}_{\mathrm{int}},$$

where $\hat{H}_{\mathrm{S}}$ and $\hat{H}_{\mathrm{E}}$ act on the individual Hilbert spaces $\mathcal{H}_{\mathrm{S}}$ and $\mathcal{H}_{\mathrm{E}}$, respectively, while $\hat{H}_{\mathrm{int}}$ acts on the composite space $\mathcal{H}_{\mathrm{T}}$ and describes the system–environment interaction. The dimensionless parameter α is the coupling strength parameter.

The open-system setup is sketched in Fig. 2.1.

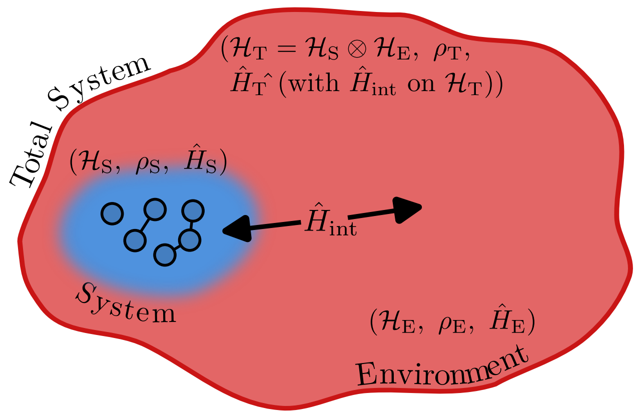

Figure 2.1: Schematic open-system setup. A system S is coupled to an environment E via $\hat{H}_{\mathrm{int}}$. Schematically the system is indicated to contain six interacting particles such as atoms or molecules, while the environment has infinitely many degrees of freedom.

The interaction Hamiltonian is typically written in the form

$$\hat{H}_{\mathrm{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i, \tag{2.6}$$

where $\hat{S}_i$ are system operators and $\hat{E}_i$ are environment operators. These will be concretized later in Sec. 2.3.

2.1.3 Interaction Picture

To describe the system dynamics, the interaction picture is used, where operators evolve with respect to the free Hamiltonian $\hat{H}_0 = \hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{E}}$. Any operator $\hat{O}$ in the Schrödinger picture takes the form

$$\hat{O}_{\mathrm{I}}(t) = e^{\mathrm{i}(\hat{H}_0)t}\, \hat{O}\, e^{-\mathrm{i}(\hat{H}_0)t},$$

in the interaction picture. Throughout this thesis, hats indicate operators, while the subscript I indicates an interaction-picture quantity.

This transformation isolates the effect of the system-bath interaction $\hat{H}_{\mathrm{int}}$, allowing systematic perturbative treatment of weak coupling. The same interaction-picture machinery will later be applied to the light–matter interaction in spectroscopy (see Ch. 3 and App. B), but at the level of the already-reduced system.

States now evolve only according to the interaction Hamiltonian $\hat{H}_{\mathrm{int}}$, and Eq. (2.5) becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{T,I}}(t) = -\mathrm{i}\alpha[\hat{H}_{\mathrm{int,I}}(t), \hat{\rho}_{\mathrm{T,I}}(t)], \tag{2.7}$$

which can be formally integrated as

$$\hat{\rho}_{\mathrm{T,I}}(t) = \hat{\rho}_{\mathrm{T,I}}(0) - \mathrm{i}\alpha \int_0^t \mathrm{d}s\, [\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(s)].$$

Inserting this back into Eq. (2.7) yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{T,I}}(t) = -\mathrm{i}\alpha \left[\hat{H}_{\mathrm{int,I}}(t), \hat{\rho}_{\mathrm{T,I}}(0)\right] - \alpha^2 \int_0^t \left[\hat{H}_{\mathrm{int,I}}(t), \left[\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(s)\right]\right]\mathrm{d}s,$$

The iteration can be repeated, leading to a series expansion in powers of α

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{T,I}}(t) = -\mathrm{i}\alpha \left[\hat{H}_{\mathrm{int,I}}(t), \hat{\rho}_{\mathrm{T,I}}(0)\right] - \alpha^2 \int_0^t \left[\hat{H}_{\mathrm{int,I}}(t), \left[\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(s)\right]\right]\mathrm{d}s + \mathcal{O}(\alpha^3). \tag{2.8}$$

The expansion is truncated at second order, justified by the weak coupling assumption $\alpha \ll 1$, which represents the **Born approximation**.

Equation (2.8) still contains the full history through $\hat{\rho}_{\mathrm{T,I}}(s)$ inside the integral and is therefore *non-Markovian*. As a first step toward a time-local Born–Markov description, one assumes that the total state varies slowly on the bath correlation time and replaces $\hat{\rho}_{\mathrm{T,I}}(s) \to \hat{\rho}_{\mathrm{T,I}}(t)$ inside the integrand. This is a time-local substitution at second order and yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{T,I}}(t) = -\mathrm{i}\alpha\big[\hat{H}_{\mathrm{int,I}}(t), \hat{\rho}_{\mathrm{T,I}}(0)\big] - \alpha^2 \int_0^t \mathrm{d}s\, \big[\hat{H}_{\mathrm{int,I}}(t), [\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(t)]\big], \tag{2.9}$$

which is now local in $\hat{\rho}_{\mathrm{T,I}}(t)$ (but still retains an explicit upper integration limit t so is not yet Markovian). The remaining Markov step is the extension of the upper limit to ∞ using short bath memory. The exact equation of motion for the reduced state $\hat{\rho}(t)$ involves the full many-body dynamics of the environment, which is generally intractable. Therefore, the next step is to trace out the environmental degrees of freedom.

2.1.4 Reduced Dynamics and Born Approximation

Since the dynamics of the system alone is of interest, the reduced density matrix is defined using the partial trace operation in Eq. (2.3):

$$\hat{\rho}_{\mathrm{S,I}}(t) = \mathrm{Tr}_{\mathrm{E}}[\hat{\rho}_{\mathrm{T,I}}(t)].$$

Thus Eq. (2.9) becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S,I}}(t) = -\mathrm{i}\,\alpha\, \mathrm{Tr}_{\mathrm{E}}\big[\, \hat{H}_{\mathrm{int,I}}(t), \hat{\rho}_{\mathrm{T,I}}(0)\,\big] - \alpha^2 \int_0^t \mathrm{d}s\, \mathrm{Tr}_{\mathrm{E}}\big[\, \hat{H}_{\mathrm{int},I}(t), [\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(t)]\,\big]$$

Vanishing First-Order Term

The first order term in α in Eq. (2.10) will vanish after tracing over the environment if the interaction operators satisfy

$$\langle \hat{E}_i \rangle_0 \equiv \mathrm{Tr}_{\mathrm{E}}[\hat{E}_i\, \hat{\rho}_{\mathrm{E}}(0)] = 0, \tag{2.10}$$

This is achieved by assuming that the total system starts in a **separable** product state of the form

$$\hat{\rho}_{\mathrm{T,I}}(0) = \hat{\rho}_{\mathrm{S,I}}(0) \otimes \hat{\rho}_{\mathrm{E}}(0), \tag{2.11}$$

Some textbooks impose the condition Eq. (2.10) as a starting assumption. Here a step further is taken. When it is not initially satisfied, it can *always* be enforced by redefining the Hamiltonian through a shift

$$\hat{H}_{\mathrm{T}} = \hat{H}'_{\mathrm{S}} + \hat{H}_{\mathrm{E}} + \alpha \hat{H}'_{\mathrm{int}},$$

with

$$\hat{H}'_{\text{int}} = \sum_i \hat{S}_i \otimes \hat{E}'_i, \qquad \hat{E}'_i = \hat{E}_i - \langle \hat{E}_i \rangle_0, \qquad \langle \hat{E}_i \rangle_0 \equiv \mathrm{Tr}_{\mathrm{E}}[\hat{E}_i\, \hat{\rho}_{\mathrm{E}}(0)],$$

and a correspondingly shifted system Hamiltonian

$$\hat{H}'_{\mathrm{S}} = \hat{H}_{\mathrm{S}} + \alpha \sum_i \hat{S}_i \langle \hat{E}_i \rangle_0.$$

This “renormalization” merely redefines system energy levels and does not affect dissipative structure. Hence, without loss of generality the shift is assumed to be performed so that $\langle \hat{E}_i \rangle_0 = 0$. Explicitly, the first-order term in Eq. (2.10) vanishes:

$$\sum_i \mathrm{Tr}_{\mathrm{E}}\left[\hat{S}_i \otimes \hat{E}_i, \hat{\rho}_{\mathrm{S,I}}(0) \otimes \hat{\rho}_{\mathrm{E}}(0)\right] = \sum_i \left(\hat{S}_i \hat{\rho}_{\mathrm{S,I}}(0) - \hat{\rho}_{\mathrm{S,I}}(0)\hat{S}_i\right) \mathrm{Tr}_{\mathrm{E}}[\hat{E}_i\, \hat{\rho}_{\mathrm{E}}(0)] = 0,$$

which implements that a static bath mean field can be absorbed into $\hat{H}_{\mathrm{S}}$. The reduced equation of motion becomes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S,I}}(t) &= -\mathrm{i}\,\alpha\, \mathrm{Tr}_{\mathrm{E}}\left[\, \hat{H}_{\mathrm{int,I}}(t), \hat{\rho}_{\mathrm{T,I}}(0)\,\right] - \alpha^2 \int_0^t \mathrm{d}s\, \mathrm{Tr}_{\mathrm{E}}\left[\, \hat{H}_{\mathrm{int,I}}(t), [\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(t)]\,\right] \\ &= -\alpha^2 \int_0^t \mathrm{d}s\, \mathrm{Tr}_{\mathrm{E}}\left[\, \hat{H}_{\mathrm{int,I}}(t), [\hat{H}_{\mathrm{int,I}}(s), \hat{\rho}_{\mathrm{T,I}}(t)]\,\right]. \end{aligned} \tag{2.12}$$

To make the time-translation structure explicit, set $s' = t - s$ (equivalently, $\mathrm{d}s' = -\mathrm{d}s$). When s runs from 0 to t, s' runs from t to 0. Reversing the limits removes the minus sign, so the integration range remains $[0, t]$. Using this change and expanding the double commutator,

$$\big[A, [B, X]\big] = ABX - AXB - BXA + XBA,$$

Eq. (2.12) can be rewritten in the form

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S,I}}(t) = \alpha^2 \int_0^t \mathrm{d}s\, \mathrm{Tr}_{\mathrm{E}}\Big\{ &\hat{H}_{\mathrm{int,I}}(t)\big[\hat{H}_{\mathrm{int,I}}(t-s)\hat{\rho}_{\mathrm{T,I}}(t) - \hat{\rho}_{\mathrm{T,I}}(t)\,\hat{H}_{\mathrm{int,I}}(t-s)\big] \\ &- \big[\hat{H}_{\mathrm{int,I}}(t-s)\hat{\rho}_{\mathrm{T,I}}(t) - \hat{\rho}_{\mathrm{T,I}}(t)\,\hat{H}_{\mathrm{int,I}}(t-s)\big]\hat{H}_{\mathrm{int,I}}(t)\Big\}. \end{aligned}$$

Next, the assumption of Eq. (2.11) is strengthened by requiring that throughout the evolution the total state remains close to a factorized form $\hat{\rho}_{\mathrm{T,I}}(t) \approx \hat{\rho}_{\mathrm{S,I}}(t) \otimes \hat{\rho}_{\mathrm{E,I}}(t)$. This can again be stated as the **Born approximation** of weak coupling. Collecting all system operators and all environment operators, and inserting the interaction Hamiltonian Eq. (2.6) explicitly, the following expression is obtained. Operators at time $t - s$ are tracked with index i' and at time t with i

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S,I}}(t) = \alpha^2 \sum_{i,i'} \int_0^t \mathrm{d}s \Big\{ &\mathrm{Tr}_{\mathrm{E}}\big[\hat{S}_{i,\mathrm{I}}(t)\hat{S}_{i',\mathrm{I}}(t-s)\hat{\rho}_{\mathrm{S,I}}(t) \otimes \hat{E}_{i,\mathrm{I}}(t)\hat{E}_{i',\mathrm{I}}(t-s)\hat{\rho}_{\mathrm{E,I}}(t)\big] - \\ &\mathrm{Tr}_{\mathrm{E}}\big[\hat{S}_{i,\mathrm{I}}(t)\hat{\rho}_{\mathrm{S,I}}(t)\hat{S}_{i',\mathrm{I}}(t-s) \otimes \hat{E}_{i,\mathrm{I}}(t)\hat{\rho}_{\mathrm{E,I}}(t)\hat{E}_{i',\mathrm{I}}(t-s)\big] - \\ &\mathrm{Tr}_{\mathrm{E}}\big[\hat{S}_{i',\mathrm{I}}(t-s)\hat{\rho}_{\mathrm{S,I}}(t)\hat{S}_{i,\mathrm{I}}(t) \otimes \hat{E}_{i',\mathrm{I}}(t-s)\hat{\rho}_{\mathrm{E,I}}(t)\hat{E}_{i,\mathrm{I}}(t)\big] + \\ &\mathrm{Tr}_{\mathrm{E}}\big[\hat{\rho}_{\mathrm{S,I}}(t)\hat{S}_{i',\mathrm{I}}(t-s)\hat{S}_{i,\mathrm{I}}(t) \otimes \hat{\rho}_{\mathrm{E,I}}(t)\hat{E}_{i',\mathrm{I}}(t-s)\hat{E}_{i,\mathrm{I}}(t)\big]\Big\}. \end{aligned}$$

Since the trace only acts on the environment, the system operators can be taken out of the trace, and the two-point correlation functions are defined using Eq. (2.4)

$$C_{ii'}(t, s) \equiv \langle \hat{E}_{i,\mathrm{I}}(t)\hat{E}_{i',\mathrm{I}}(t-s)\rangle = \mathrm{Tr}_{\mathrm{E}}\big[\hat{E}_{i,\mathrm{I}}(t)\hat{E}_{i',\mathrm{I}}(t-s)\,\hat{\rho}_{\mathrm{E,I}}(t)\big]. \tag{2.13}$$

The term 'two-point' indicates that the bath operators $\hat{E}_i$ and $\hat{E}_{i'}$ are evaluated at two different times. Up to this point, the derivation has been general and no assumption of stationarity was required. The correlator can depend on both times through $\hat{\rho}_{\mathrm{E,I}}(t)$. Now the bath is taken to be stationary, i.e. $[\hat{H}_{\mathrm{E}}, \hat{\rho}_{\mathrm{E}}(0)] = 0$. This implies $\hat{\rho}_{\mathrm{E,I}}(t) = \hat{\rho}_{\mathrm{E}}(0)$ and therefore

$$C_{ii'}(t,s) = C_{ii'}(t-s).$$

To make this dependence explicit we introduce the time difference

$$\tau \equiv t - s,$$

and subsequently write $C_{ii'}(\tau)$ in place of $C_{ii'}(t,s)$.

Units. With $\hat{H}_{\mathrm{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i$ and $\hbar = 1$, $\hat{H}_{\mathrm{int}}$ has units of time^{-1}, hence $[\hat{S}_i][\hat{E}_i] = \text{time}^{-1}$. If $\hat{S}_i$ is chosen dimensionless, then $[\hat{E}_i] = \text{time}^{-1}$ and therefore $[C_{ii'}] = \text{time}^{-2}$.

If the bath operators are Hermitian, i.e. $\hat{E}_i^\dagger = \hat{E}_i$, and the bath state is stationary (i.e., $[\hat{H}_{\mathrm{E}}, \hat{\rho}_{\mathrm{E}}(0)] = 0$), the cyclic property of the trace gives the conjugation relation

$$C^*_{ii'}(\tau) = \tilde{C}_{ii'}(\tau) \equiv \langle \hat{E}_{i',\mathrm{I}}(t-\tau)\hat{E}_{i,\mathrm{I}}(t)\rangle,$$

In the interaction picture, stationarity implies $\hat{\rho}_{\mathrm{E,I}}(t) = \hat{\rho}_{\mathrm{E,I}}(0) = \hat{\rho}_{\mathrm{E}}(0)$ and the correlators depend only on the time difference $\tau = t - s$. With $\hat{E}_{i,\mathrm{I}}(t) = e^{\mathrm{i}\hat{H}_{\mathrm{E}}t}\hat{E}_i e^{-\mathrm{i}\hat{H}_{\mathrm{E}}t}$ this implies the further symmetry condition

$$C^*_{ii'}(-\tau) = C_{ii'}(\tau).$$

and thus finally obtain the desired form of the Redfield equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S,I}}(t) = \alpha^2 \sum_{i,i'} \int_0^t \mathrm{d}s \Big\{ C_{ii'}(t-s)\big[\hat{S}_{i,\mathrm{I}}(t), \hat{S}_{i',\mathrm{I}}(t-s)\hat{\rho}_{\mathrm{S,I}}(t)\big] + \mathrm{H.c.} \Big\}. \tag{2.14}$$

This form shows explicitly that the system evolution depends only on bath correlation functions evaluated along the free bath dynamics.

2.1.5 Markov Approximation

However Eq. (2.14) is still not Markovian, since it carries an explicit integration over time t. An additional approximation is therefore introduced. Besides weak coupling, assume that the bath correlation functions decay on a characteristic time scale τ_E that is much shorter than the relevant system evolution time τ_S, i.e. $\tau_E \ll \tau_S$. In particular, $C_{ii'}(\tau)$ becomes negligible for $\tau \gtrsim \tau_E$, so the integrand has support only on a short memory window. If the reduced state varies slowly on this window, one may approximate it as constant over the integration range and extend the upper limit to infinity, which yields a Markovian generator for $t \gg \tau_E$.

$$\int_0^t \mathrm{d}s\, C_{ii'}(t-s) f(s) \;\longrightarrow\; \int_0^\infty \mathrm{d}\tau\, C_{ii'}(\tau) f(t), \qquad (t \gg \tau_E),$$

where the slow variation of $\hat{\rho}_{\mathrm{S,I}}(t)$ and of the system operators on the time scale τ_E has been used. Applying this rule to Eq. (2.12) yields a *time-local* Markovian Redfield generator. In the interaction picture the operators still carry explicit t-dependence, but the kernel is time-translation invariant and depends only on τ, so the generator has no explicit dependence on absolute time beyond the system operators themselves

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S,I}}(t) = -\alpha^2 \sum_{i,i'} \int_0^\infty \mathrm{d}\tau \Big(C_{ii'}(\tau)[\hat{S}_{i,\mathrm{I}}(t), \hat{S}_{i',\mathrm{I}}(t-\tau)\hat{\rho}_{\mathrm{S,I}}(t)] + \mathrm{H.c.} \Big). \tag{2.15}$$

Here $\hat{S}_{i,\mathrm{I}}(t-\tau)$ denotes the free interaction-picture evolution under $\hat{H}_{\mathrm{S}}$, so the explicit $(t-\tau)$ dependence reflects free system dynamics rather than any memory in the kernel. This step encapsulates the loss of memory: the generator now includes a very short time window τ_E where the system operators can be approximated as constant (frozen at time $\hat{\rho}_{\mathrm{S,I}}(t)$). Equation Eq. (2.15) is the Redfield master equation in the interaction picture under the Born (weak-coupling) and Markov (short-memory) approximations. In this form, all environmental effects are encoded in the bath correlation functions $C_{ii'}(\tau)$, which are analyzed in the next section.

2.2 Environmental Correlators and Spectral Properties

In what follows, the origin and key properties of bath correlation functions (defined by Eq. (2.13)) are outlined, their spectral Fourier representation is introduced, and the simultaneous encoding of emission and absorption processes is discussed. In particular, the decay of $C_{ii'}(\tau)$ on the bath correlation time τ_E quantifies how rapidly the environment "forgets", and the Markov approximation used above is justified when τ_E is much shorter than the relevant system evolution time scales. The connection between $C_{ii'}(\tau)$ and the spectral density is direct and experimentally accessible. Modern noise spectroscopy techniques enable direct measurement of bath spectral properties from coherence-decay observations without requiring pulse sequences or prior knowledge of the system-bath coupling constants—providing parameter-free experimental access to the bath structure that drives decoherence in the e.g. molecular systems [23].

2.2.1 Fourier Representation and Secular Approximation

For a stationary thermal bath in equilibrium, where its state is given by the Gibbs state (see Sec. 2.3.2), the Kubo–Martin–Schwinger (KMS) condition relates correlations at positive and negative times via a shift into the complex time plane,

$$C_{ii'}(\tau) \equiv \langle \hat{E}_{i,\mathrm{I}}(\tau)\hat{E}_{i',\mathrm{I}}(0)\rangle = \langle \hat{E}_{i,\mathrm{I}}(0)\hat{E}_{i',\mathrm{I}}(\tau + \mathrm{i}\beta)\rangle \equiv C_{i'i}(+\tau + \mathrm{i}\beta). \tag{2.16}$$

A short proof of this relation in the eigenbasis of the bath Hamiltonian is as follows

$$\begin{aligned}
C_{ii'}(\tau + \mathrm{i}\beta) &= \mathrm{Tr}(\hat{\rho}_\beta\, \hat{E}_i\, \hat{E}_{i',\mathrm{I}}(\tau + \mathrm{i}\beta)) \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_n} \langle n|\, \hat{E}_i\, |m\rangle\; \langle m|\, \hat{E}_{i',\mathrm{I}}(\tau + \mathrm{i}\beta)\, |n\rangle \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_n} e^{\mathrm{i}(E_m - E_n)(\tau + \mathrm{i}\beta)} (\hat{E}_i)_{nm} (\hat{E}_{i'})_{mn} \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_m} e^{\mathrm{i}(E_n - E_m)\tau} (\hat{E}_{i'})_{nm} (\hat{E}_i)_{mn} \\
&= C_{i'i}(\tau).
\end{aligned}$$

Eigenoperator Decomposition

To proceed analytically and to connect with relaxation pathways, the system coupling operators are decomposed into eigenoperators of the system Hamiltonian. Let $\hat{H}_{\mathrm{S}} = \sum_\epsilon \epsilon\, \hat{\Pi}_\epsilon$ be the spectral resolution with projectors $\hat{\Pi}_\epsilon$. The Bohr frequencies are defined as $\omega = \epsilon' - \epsilon$ and the corresponding interaction operators in the eigenbasis are

$$\hat{S}_i(\omega) = \sum_{\epsilon' - \epsilon = \omega} \hat{\Pi}_\epsilon\, \hat{S}_i\, \hat{\Pi}_{\epsilon'}.$$

In the interaction picture these acquire simple oscillatory phases

$$\hat{S}_{i,\mathrm{I}}(t) = \sum_{\omega} e^{-\mathrm{i}\omega t} \hat{S}_i(\omega), \qquad \hat{S}_{i',\mathrm{I}}(t-\tau) = \sum_{\omega'} e^{-\mathrm{i}\omega'(t-\tau)} \hat{S}_{i'}(\omega'). \tag{2.17}$$

Note that a time-independent system Hamiltonian $\hat{H}_\mathrm{S}$ is assumed here. The derivation can in principle be extended to time-dependent $\hat{H}_\mathrm{S}(t)$, for example in spectroscopy where external driving fields lead to an explicitly time-dependent system Hamiltonian [3] (see App. B). Generalizations of Markovian master-equation derivations to driven systems are discussed in [24].

Inserting Eq. (2.17) into Eq. (2.15) produces sums over oscillatory factors $e^{-\mathrm{i}(\omega-\omega')t}$ multiplying integrals of $C_{ii'}(\tau)e^{\mathrm{i}\omega'\tau}$. The inner commutator in Eq. (2.15) becomes

$$[\hat{S}_{i,\mathrm{I}}(t),\, \hat{S}_{i',\mathrm{I}}(t-\tau)\, \hat{\rho}_{\mathrm{S},\mathrm{I}}(t)] = \sum_{\omega,\omega'} e^{-\mathrm{i}(\omega-\omega')t} e^{+\mathrm{i}\omega'\tau} \big[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{\mathrm{S},\mathrm{I}}(t)\big],$$

and the total Redfield equation reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S},\mathrm{I}}(t) = -\alpha^2 \sum_{i,i'} \sum_{\omega,\omega'} e^{-\mathrm{i}(\omega-\omega')t} \left(\int_0^\infty \mathrm{d}\tau\, C_{ii'}(\tau)\, e^{+\mathrm{i}\omega'\tau} \right) \big[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{\mathrm{S},\mathrm{I}}(t)\big] + \mathrm{H.c.}. \tag{2.18}$$

The correlator is now expressed in the frequency domain and the noise-power spectrum of the environment is defined [25]. To avoid confusion with the system coupling operators $\hat{S}_i$, we denote the bath spectrum by $\mathcal{S}(\omega)$ (common in the literature as $S(\omega)$).

$$\mathcal{S}(\omega) \equiv \int_{-\infty}^{\infty} \mathrm{d}\tau\, C(\tau)\, e^{+\mathrm{i}\omega\tau}, \tag{2.19}$$

Units. Since $[C] = \mathrm{time}^{-2}$ for dimensionless $\hat{S}_i$ and $\hbar = 1$, the Fourier transform implies $[\mathcal{S}(\omega)] = [C]\,\mathrm{time} = \mathrm{time}^{-1}$.

This quantity tells how strongly bath fluctuations at frequency ω contribute to noise. For a stationary bath correlation function $C(\tau)$, the noise spectrum $\mathcal{S}(\omega)$ is by the Wiener–Khinchin relation the Fourier transform of $C(\tau)$ and real and non-negative $\mathcal{S}(\omega) \geq 0$ [26].

The KMS condition Eq. (2.16) can now be translated to frequency domain, where it imposes the detailed-balance symmetry

$$\mathcal{S}(-\omega) = e^{-\omega\beta}\mathcal{S}(\omega), \tag{2.20}$$

which ensures that upward and downward transition rates obey Boltzmann ratios.

The exact one-sided Fourier (Laplace-) transform of the bath correlation function can be split into the power spectrum and an energy shift $\lambda(\omega)$ as

$$\begin{aligned} \chi_{ii'}(\omega) &\equiv \int_0^\infty \mathrm{d}\tau\, C_{ii'}(\tau)\, e^{+\mathrm{i}\omega\tau} \\ &= \tfrac{1}{2}\mathcal{S}_{ii'}(\omega) + \mathrm{i}\,\lambda_{ii'}(\omega), \end{aligned}$$

with

$$\mathcal{S}_{ii'}(\omega) = \chi_{ii'}(\omega) + \chi^*_{i'i}(\omega).$$

which reduce **??** to the compact form

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S},\mathrm{I}}(t) = -\alpha^2 \sum_{i,i'} \sum_{\omega,\omega'} e^{-\mathrm{i}(\omega-\omega')t} \chi_{ii'}(\omega') \big[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{\mathrm{S},\mathrm{I}}(t)\big] + \mathrm{H.c.}.$$

Going back to the Schrödinger picture, this yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S}}(t) = -\mathrm{i}[\hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{LS}}, \hat{\rho}_{\mathrm{S}}(t)] + \mathcal{D}_{\mathrm{R}}[\hat{\rho}_{\mathrm{S}}(t)],$$

with the Lamb-shift Hamiltonian

$$\hat{H}_{\mathrm{LS}} = \sum_{\omega}\sum_{i,i'} \lambda_{ii'}(\omega)\, \hat{S}_i^{\dagger}(\omega)\hat{S}_{i'}(\omega),$$

which will be absorbed into a renormalized system Hamiltonian and disregarded in the subsequent derivation. The Redfield dissipator is

$$\mathcal{D}_{\mathrm{R}}[\hat{\rho}] = \frac{1}{2}\sum_{\omega,\omega'}\sum_{i,i'} e^{-\mathrm{i}(\omega-\omega')t}\mathcal{S}_{ii'}(\omega')\left(\hat{S}_{i'}(\omega')\,\hat{\rho}\,\hat{S}_i^{\dagger}(\omega) - \hat{S}_i^{\dagger}(\omega)\hat{S}_{i'}(\omega')\,\hat{\rho}\right) + \mathrm{H.c.}. \tag{2.21}$$

Matrix Elements and Redfield Tensor

In the eigenbasis of the Hamiltonian $\{|a\rangle\}$, taking $\langle a|\cdot|b\rangle$ of the master equation and grouping terms gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_{ab}(t) = -\mathrm{i}\,\omega_{ab}\rho_{ab}(t) + \sum_{c,d} R_{abcd}\,\rho_{cd}(t),$$

where the Bloch–Redfield tensor, also implemented in QuTiP, is defined as

$$\begin{aligned} R_{abcd} = -\frac{1}{2}\sum_{i,i'}\Big\{&\delta_{bd}\sum_n S^i_{an}S^{i'}_{nc}\,\mathcal{S}_{ii'}(\omega_{cn}) - S^{i'}_{ac}S^i_{db}\,\mathcal{S}_{ii'}(\omega_{ca}) \\ &+ \delta_{ac}\sum_n S^i_{dn}S^{i'}_{nb}\,\mathcal{S}_{ii'}(\omega_{dn}) - S^{i'}_{ac}S^i_{db}\,\mathcal{S}_{ii'}(\omega_{db})\Big\}. \end{aligned} \tag{2.22}$$

with S^i the system coupling operators and $\mathcal{S}_{ii'}(\omega)$ the bath spectra[25].

Secular Approximation

The oscillatory prefactors $e^{-\mathrm{i}(\omega-\omega')t}$ in Eq. (2.20) average to zero on coarse-grained times Δt satisfying $\tau_E \ll \Delta t \ll 1/|\omega-\omega'|$ whenever $\omega \neq \omega'$. Keeping only terms with $\omega_{ab} = \omega_{cd}$ in Eq. (2.21) removes such fast-rotating couplings and ensures a block-diagonal structure in the Redfield tensor R_{abcd}.

This yields the *secular* Redfield equation in Lindblad form

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathrm{S}}(t) &= -\mathrm{i}[\hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{LS}}, \hat{\rho}_{\mathrm{S}}(t)] \\ &\quad + \sum_{i,i'}\sum_{\omega}\gamma_{ii'}(\omega)\Big(\hat{S}_{i'}(\omega)\hat{\rho}_{\mathrm{S}}(t)\hat{S}_i^{\dagger}(\omega) - \tfrac{1}{2}\{\hat{S}_i^{\dagger}(\omega)\hat{S}_{i'}(\omega), \hat{\rho}_{\mathrm{S}}(t)\}\Big), \end{aligned}$$

where $\gamma_{ii'}(\omega) = 2\mathrm{Re}\,\chi_{ii'}(\omega)$. This form is guaranteed to preserve complete positivity provided the matrix $[\gamma_{ii'}(\omega)]_{i,i'}$ is positive semidefinite for each frequency ω. Without the secular approximation, Eq. (2.20) need not generate a completely positive dynamical map, explaining the caution required when applying the full Redfield equation.

For a thermal bath in equilibrium, the bath correlation functions satisfy the KMS condition. In the weak-coupling limit with a full secular approximation, the resulting Lindblad generator has the Gibbs state as a stationary state (see Eq. (2.24)) [27, 28]

$$\mathcal{L}_{\mathrm{sec}}[\hat{\rho}_{\mathrm{Gibbs}}] = 0.$$

By contrast, the non-secular (Bloch-)Redfield generator retains terms with $\omega \neq \omega'$, which couple populations and coherences in the energy basis. Without full secularization, quantum detailed balance and Gibbs stationarity are not guaranteed, even for a single thermal bath satisfying KMS [1, 29]. In weak coupling, the stationary state typically differs from $\hat{\rho}_{\text{Gibbs}}$ by corrections of second order in the system–bath coupling (and may exhibit bath-induced population shifts and coherences).

An important exception is pure dephasing, where the coupling operator commutes with the system Hamiltonian, $[\hat{A}, \hat{H}_{\text{S}}] = 0$. In that case there is no energy exchange and $\hat{\rho}_\beta$ remains stationary also without secularization [1].

Uncorrelated Hermitian Couplings

If the baths are uncorrelated and $S_i = S_i^\dagger$:

$$\begin{aligned}
\mathcal{S}_{ii'}(\omega) &= \delta_{ii'}\mathcal{S}_i(\omega) \implies \\
R_{abcd} &= -\frac{1}{2}\sum_i \Big\{ \delta_{bd}\sum_n S^i_{an}S^i_{nc}\mathcal{S}_i(\omega_{cn}) - S^i_{ac}S^i_{db}\mathcal{S}_i(\omega_{ca}) \\
&\qquad + \delta_{ac}\sum_n S^i_{dn}S^i_{nb}\mathcal{S}_i(\omega_{dn}) - S^i_{ac}S^i_{db}\mathcal{S}_i(\omega_{db}) \Big\}.
\end{aligned}$$

2.2.2 Physical Interpretation: Emission and Absorption

Emission and absorption are both automatically included because $\mathcal{S}_{ii'}(\omega)$ contains positive- and negative-frequency components related by Eq. (2.19). Positive frequencies ($\omega > 0$) describe system energy loss (emission), while negative frequencies ($\omega < 0$) describe system energy gain (absorption). Detailed balance follows immediately: the ratio of absorption to emission contributions at frequency $\omega > 0$ is $e^{-\omega\beta}$. In the high-temperature limit ($\beta \to 0$), this simplifies to $\mathcal{S}(-\omega) \approx \mathcal{S}(\omega)$, making the power spectrum symmetric around $\omega = 0$ and rendering emission and absorption rates nearly equal.

The explicit relation between $J(\omega)$, the correlator, and the full spectrum is derived in the next section (see Eq. (2.32)).

2.3 Harmonic Oscillator Baths and Explicit Correlation Functions

The previous section established that the reduced dynamics is governed by the bath correlation functions $C_{ij}(\tau)$. To make the Redfield equation practical, these correlators must be evaluated for a concrete microscopic model. Here the focus is on the harmonic-oscillator bath, a widely used model that captures electromagnetic field fluctuations, phonons, and other bosonic environments.

The task now is to calculate the bath correlation function $C(\tau)$ for a concrete model. Recall that $C(\tau) = \langle E(t+\tau)E(t)\rangle$ measures how bath fluctuations at two different times are correlated in thermal equilibrium. For the harmonic oscillator bath, these correlators can be computed exactly, revealing how $C(\tau)$ connects to experimentally relevant spectral densities and governs the system's approach to thermal equilibrium.

For these explicit calculations, the following mathematical result is useful.

2.3.1 Geometric Series

The infinite geometric series

$$\mathcal{G} = a + ar + ar^2 + ar^3 + \cdots = \sum_{n=0}^{\infty} ar^n,$$

converges to

$$\mathcal{G} = \frac{a}{1-r} \quad \text{for } |r| < 1. \tag{2.23}$$

Differentiating Eq. (2.22) with respect to r gives

$$\sum_{n=0}^{\infty} n r^n = \frac{r}{(1-r)^2}, \quad |r| < 1. \tag{2.24}$$

The formal correlation functions introduced above are now specialized to the paradigmatic bosonic bath. This leads naturally to spectral-density representations and the standard phenomenological models such as Ohmic, sub-/super-Ohmic, and Drude–Lorentz.

2.3.2 Bosonic Environment and Gibbs State

A bosonic bath is modeled as a (large) collection of independent harmonic oscillators in thermal equilibrium:

The thermal state of the bosonic bath is given by the Gibbs density matrix

$$\hat{\rho}_{\text{Gibbs}} = \frac{e^{-\beta \hat{H}_{\text{E}}}}{\text{Tr}[e^{-\beta \hat{H}_{\text{E}}}]}, \tag{2.25}$$

so that for any bath operator $\hat{A}$,

$$\langle \hat{A} \rangle = \text{Tr}(\hat{\rho}_{\text{Gibbs}}\, \hat{A}) = \frac{1}{Z} \sum_n e^{-\beta E_n} A_{nn},$$

in the energy eigenbasis of $\hat{H}_{\text{E}}$, where $Z = \text{Tr}[e^{-\beta \hat{H}_{\text{E}}}]$ and $\beta = 1/(k_B T)$ is the inverse temperature of the environment. The bath Hamiltonian consists of independent harmonic oscillators

$$\hat{H}_{\text{E}} = \sum_k \omega_k \left(\hat{b}_k^\dagger \hat{b}_k + \tfrac{1}{2} \right), \tag{2.26}$$

and the average energy of mode k is

$$E_k = \omega_k (n_k + \tfrac{1}{2}),$$

with $n_k = \langle \hat{b}_k^\dagger \hat{b}_k \rangle$ the Bose–Einstein occupation number.

The zero-point contribution $E_{\text{vac}} = \frac{1}{2} \sum_k \omega_k$ adds only a mode-independent constant to the bath energy. Constant energy offsets commute with all observables, so they neither modify thermal populations nor enter the correlation functions that drive the system dynamics. This term is therefore dropped in the dynamical equations (standard for normal ordering, where raising operators are placed to the left of lowering operators). For completeness, the $\frac{1}{2}$ is retained in the partition function Z_k for normalization, but it cancels out of all correlation functions and rates. In other words, only energy *differences* influence the evolution.

Single Mode Partition Function and Occupation

For a single mode k, the thermal state reads

$$\hat{\rho} = \frac{e^{-\beta \omega_k \hat{b}_k^\dagger \hat{b}_k}}{Z_k},$$

with partition function

$$
\begin{aligned}
Z_k \equiv \mathrm{Tr}\left[e^{-\beta \hat{H}_{\mathrm{E}}}\right] &= \sum_{m=0}^{\infty} \langle m| \, e^{-\beta\omega_k(\hat{b}_k^\dagger \hat{b}_k + \frac{1}{2})} \, |m\rangle \\
&= \frac{e^{-\beta\omega_k/2}}{1-e^{-\beta\omega_k}}.
\end{aligned}
$$

The Bose–Einstein average occupation number follows using Eq. (2.23) and is given by

$$
\begin{aligned}
n_k = \langle \hat{b}_k^\dagger \hat{b}_k \rangle_{\mathrm{th}} &= \mathrm{Tr}\left[\hat{b}_k^\dagger \hat{b}_k \frac{e^{-\beta \hat{H}_{\mathrm{E}}}}{Z_k}\right] \\
&= \frac{\mathrm{Tr}\left[\hat{b}_k^\dagger \hat{b}_k e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k}\right]}{Z_k} \\
&= \frac{\sum_{m=0}^{\infty} \langle m| \, \hat{b}_k^\dagger \hat{b}_k e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k} \, |m\rangle}{\frac{e^{-\beta\omega_k/2}}{1-e^{-\beta\omega_k}}} \\
&= \frac{e^{-\beta\omega_k}}{1-e^{-\beta\omega_k}} \\
&= \frac{1}{e^{\beta\omega_k}-1}.
\end{aligned}
\tag{2.27}
$$

Since the bath is composed of many independent modes, this calculation can be done for every mode, and the total partition function factorizes to

$$
Z_{\mathrm{bath}} = \prod_k Z_k = \prod_k \frac{e^{-\beta\omega_k/2}}{1-e^{-\beta\omega_k}}.
$$

With the thermal properties of the bath established, attention can be turned to its coupling to a small system of interest. The bath correlation functions from Eq. (2.13) are therefore calculated for the simple case of a single coupling operator $\hat{E}$. The derivation follows and slightly expands [30].

2.3.3 Microscopic Form of the Bath Correlator

With linear system–bath coupling to oscillator displacements, the Caldeira–Leggett model [1, 31] gives the single bath coupling operator as

$$
\hat{E} = \sum_{n=1}^{\infty} c_n \hat{x}_n, \qquad \hat{x}_n = \sqrt{\frac{1}{2m_n\omega_n}}(\hat{b}_n + \hat{b}_n^\dagger).
$$

In the interaction picture the bath operators evolve as $\hat{E}_{\mathrm{I}}(t) = e^{\mathrm{i}\hat{H}_{\mathrm{E}}t}\hat{E}e^{-\mathrm{i}\hat{H}_{\mathrm{E}}t}$ with $\hat{H}_{\mathrm{E}}$ from Eq. (2.25), yielding

$$
\hat{E}_{\mathrm{I}}(0) = \sum_n c_n \sqrt{\frac{1}{2m_n\omega_n}}(\hat{b}_n + \hat{b}_n^\dagger), \tag{2.28}
$$

$$
\hat{E}_{\mathrm{I}}(\tau) = \sum_n c_n \sqrt{\frac{1}{2m_n\omega_n}}\left(\hat{b}_n e^{-\mathrm{i}\omega_n\tau} + \hat{b}_n^\dagger e^{\mathrm{i}\omega_n\tau}\right). \tag{2.29}
$$

The thermal correlation function for this single relevant bath operator is

$$
C(\tau) = \langle \hat{E}_{\mathrm{I}}(\tau)\hat{E}_{\mathrm{I}}(0)\rangle.
$$

Inserting Eqs. (2.27) and (2.28) yields a double sum over bath modes,

$$\begin{aligned} C(\tau) &= \sum_{n,m} \frac{c_n c_m}{\sqrt{4 m_n m_m \omega_n \omega_m}} \Big\langle \big(\hat{b}_n e^{-\mathrm{i}\omega_n \tau} + \hat{b}_n^\dagger e^{\mathrm{i}\omega_n \tau}\big)(\hat{b}_m + \hat{b}_m^\dagger) \Big\rangle, \\ &= \sum_{n,m} \frac{c_n c_m}{\sqrt{4 m_n m_m \omega_n \omega_m}} \Big(e^{-\mathrm{i}\omega_n \tau} \langle \hat{b}_n \hat{b}_m^\dagger \rangle + e^{\mathrm{i}\omega_n \tau} \langle \hat{b}_n^\dagger \hat{b}_m \rangle \Big), \end{aligned}$$

where mixed terms such as $\langle \hat{b}_n \hat{b}_m \rangle$ vanish for a thermal state of uncoupled oscillators. Doing similar calculations to Eq. (2.26) gives

$$\langle \hat{b}_n \hat{b}_m^\dagger \rangle = \delta_{nm}(n_n + 1) \quad \text{and} \quad \langle \hat{b}_n^\dagger \hat{b}_m \rangle = \delta_{nm} n_n,$$

with $n_n = \langle \hat{b}_n^\dagger \hat{b}_n \rangle = 1/(e^{\beta\omega_n} - 1)$ given by Eq. (2.26). Using these results, the discrete form of the bath correlation function is obtained as

$$C(\tau) = \sum_n \frac{c_n^2}{2 m_n \omega_n} \Big[(n_n + 1) e^{-\mathrm{i}\omega_n \tau} + n_n e^{\mathrm{i}\omega_n \tau} \Big].$$

2.3.4 Spectral Density Representation

To connect this discrete expression with the continuum form, note that the correlator can be expanded by writing $e^{\pm \mathrm{i}\omega_n \tau} = \cos(\omega_n \tau) \pm \mathrm{i} \sin(\omega_n \tau)$. This yields

$$\begin{aligned} C(\tau) &= \sum_n \frac{c_n^2}{2 m_n \omega_n} \Big[(n_n + 1)(\cos(\omega_n \tau) - \mathrm{i} \sin(\omega_n \tau)) + n_n (\cos(\omega_n \tau) + \mathrm{i} \sin(\omega_n \tau)) \Big] \\ &= \sum_n \frac{c_n^2}{2 m_n \omega_n} \Big[(2 n_n + 1) \cos(\omega_n \tau) - \mathrm{i} \sin(\omega_n \tau) \Big]. \end{aligned}$$

Since $2n_n + 1 = 2/(e^{\beta\omega_n} - 1) + 1 = \coth(\beta\omega_n/2)$ (recall $\coth(x/2) = (e^x + 1)/(e^x - 1)$), this can be rewritten as

$$C(\tau) = \sum_n \frac{c_n^2}{2 m_n \omega_n} \Big[\coth\Big(\frac{\beta\omega_n}{2}\Big) \cos(\omega_n \tau) - \mathrm{i} \sin(\omega_n \tau) \Big]. \tag{2.30}$$

To move from discrete modes to a continuum description, introduce the spectral density, which encodes both the mode density and coupling strength,

$$J(\omega) = \pi \sum_n \frac{c_n^2}{2 m_n \omega_n} \delta(\omega - \omega_n),$$

which summarizes how strongly each bath frequency couples to the system. Here it is assumed that the bath has discrete modes ω_n and each couples to the system with strength $\propto c_n$. By construction, the delta-function representation implies the distributional identity that for any smooth test function $f(\omega)$,

$$\sum_n \frac{c_n^2}{2 m_n \omega_n} f(\omega_n) = \frac{1}{\pi} \int_0^\infty \mathrm{d}\omega \, J(\omega) f(\omega). \tag{2.31}$$

This simply means that the discrete mode sum can be replaced by a continuum integral once $J(\omega)$ is defined. We then choose the specific function $f(\omega) = \coth(\beta\omega/2)\cos(\omega\tau) - i\sin(\omega\tau)$ because it is exactly the factor appearing in the discrete correlator. Applying the identity with this f yields a representation of the bath correlator in a continuum limit

$$C(\tau) = \int_0^\infty \mathrm{d}\omega \, \frac{J(\omega)}{\pi} \Big[\coth\Big(\frac{\beta\omega}{2}\Big) \cos(\omega\tau) - i \sin(\omega\tau) \Big]. \tag{2.32}$$

The cosine term is even in τ and captures symmetrized noise, while the sine term is odd and encodes dissipation. In the macroscopic limit, the mode set becomes dense, and the spectral density $J(\omega)$ can be expressed as $J(\omega) = \rho(\omega)g(\omega)^2$, where $\rho(\omega)$ is the density of states and $g(\omega)$ the form factor. The spectral density thus fully characterizes the environmental coupling: it determines which bath frequencies exist and how strongly each frequency couples to the system. The real part of $C(\tau)$ describes symmetrized noise, while the imaginary part describes dissipation [1, 2, 32].

The parametric choice of $J(\omega)$, i.e., the bath spectral density, fixes whether noise is dominated by low-frequency components and whether the bath is smooth or has sharp features [1, 2]. This will be illustrated visually in Sec. 2.4, see in particular Fig. 2.2 and Fig. 2.4.

The bath correlation function $C(\tau)$ is the key quantity determining the system dynamics in the Redfield equation. It is fully specified by the spectral density $J(\omega)$ and the temperature-dependent factor $\coth(\beta\omega/2)$, which weights each frequency according to thermal occupation. The spectral density is also directly related to the noise power spectrum Eq. (2.18):

$$\mathcal{S}(\omega) = \int_{-\infty}^{\infty} \mathrm{d}\tau\, C(\tau) e^{+\mathrm{i}\omega\tau} = 2\pi J(|\omega|) \begin{cases} n_{\mathrm{th}}(\omega) + 1, & \omega > 0, \\ n_{\mathrm{th}}(|\omega|), & \omega < 0, \end{cases} \tag{2.33}$$

Here $J(\omega)$ is defined for $\omega > 0$, and the thermal occupation is $n_{\mathrm{th}}(\omega) = 1/(e^{\beta\omega} - 1)$. The piecewise form makes explicit that $\omega > 0$ corresponds to emission and $\omega < 0$ to absorption, with the detailed-balance ratio fixed by $e^{-\beta\omega}$. Thus, specifying either the correlator $C(\tau)$, the noise power spectrum $\mathcal{S}(\omega)$, or the spectral density $J(\omega)$ together with the temperature β suffices to fully characterize the environmental coupling. The spectral density is often the most convenient choice, as it is a real-valued function of a single variable and can be directly related to microscopic models.

In the following, some of the most common choices for $J(\omega)$ are introduced.

2.4 Common Choices of Spectral Densities

The main qualitative differences between bath models are set by the low-frequency scaling of $J(\omega)$ and by how high-frequency modes are regularized (cutoff). Both directly impact correlation times and the Markov regime.

2.4.1 Ohmic and Related Spectral Densities

The Redfield equation can also be connected to classical phenomenological models of dissipation. For an Ohmic spectral density, the low-frequency behavior is linear, and the spectral density is given by [2]

$$J(\omega) \propto \gamma\omega,$$

where γ is the damping constant. This corresponds to a classical friction-like dissipation. To ensure physical behavior, a cutoff is introduced

$$J(\omega) = \alpha \frac{\omega^s}{\omega_c^{s-1}} e^{-\omega/\omega_c}, \tag{2.34}$$

which enforces convergence of integrals such as Eq. (2.32). Here α is a dimensionless coupling constant, ω_c is the cutoff frequency, and s controls the low-frequency behavior of the bath, thereby effecting the timescale of the environment. The parameter s classifies the spectral density into three regimes: Ohmic ($s = 1$), *sub-Ohmic* ($s < 1$), and *super-Ohmic* ($s > 1$) [2, 25]. Sub-Ohmic baths exhibit slower fluctuations and longer memory, while super-Ohmic baths suppress low-frequency noise and emphasize fast, high-frequency modes. To make this separation explicit in the numerical illustrations below, it is convenient to measure all frequencies in units of a characteristic system scale ω_0 and to express bath parameters as dimensionless ratios (ω_c/ω_0, T/ω_0, α/ω_0). As shown in Fig. 2.2, the sub-Ohmic correlation function remains nonzero for longer times than the Ohmic and super-Ohmic cases.

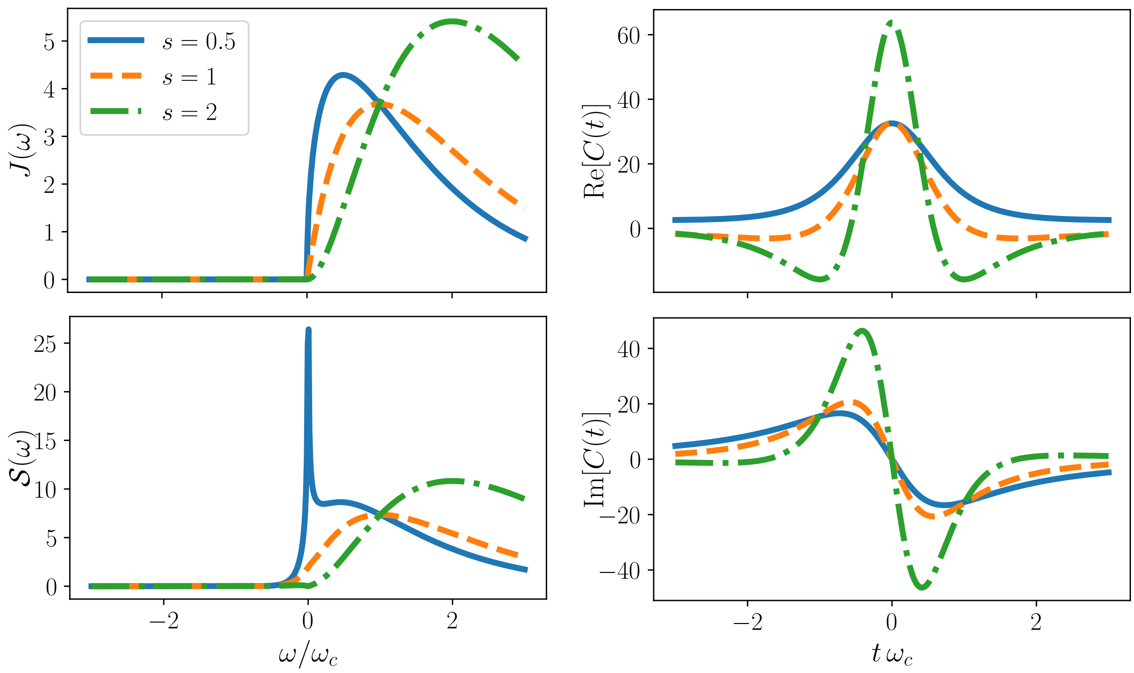

Figure 2.2: Comparison of sub-Ohmic ($s = 0.5$), Ohmic ($s = 1$), and super-Ohmic ($s = 2$) spectral densities, power spectra, and correlation functions. Parameters: $\omega_c/\omega_0 = 10$, $T/\omega_0 = 1$, $\alpha/\omega_0 = 1$.

For an Ohmic bath, increasing ω_c pushes bath weight to shorter times and reduces the bath memory time. This is precisely the physical content behind the Markov approximation and is illustrated in Fig. 2.3.

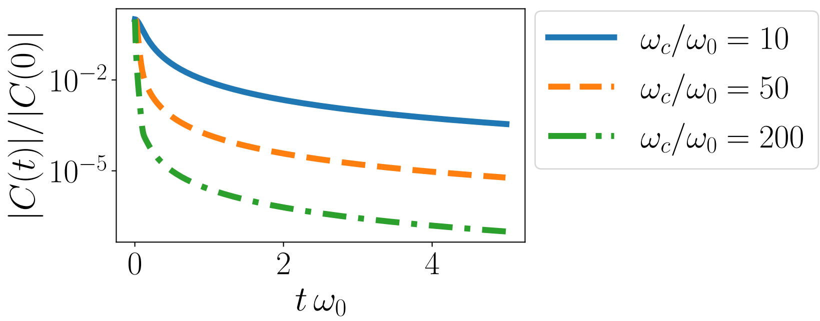

Figure 2.3: Normalized bath correlation magnitude $|C(t)|/|C(0)|$ for an Ohmic bath ($s = 1$), showing faster decay for larger cutoff ω_c/ω_0. Parameters: $T/\omega_0 = 10$, $\alpha/\omega_0 = 10^{-3}$.

Another widely used model with linear low-frequency behavior is the Drude–Lorentz spectral density, which introduces a Lorentzian cutoff

$$J(\omega) = 2\lambda \frac{\omega\gamma}{\omega^2 + \gamma^2},$$

where λ is the reorganization energy, and γ is interpreted as a cutoff. This model represents an overdamped Brownian oscillator. A visual comparison between the Ohmic and Drude–Lorentz spectral densities, together with their corresponding correlation functions and power spectra, is shown in Fig. 2.4. The Drude–Lorentz model exhibits a longer high-frequency tail than the exponentially cut off Ohmic case.

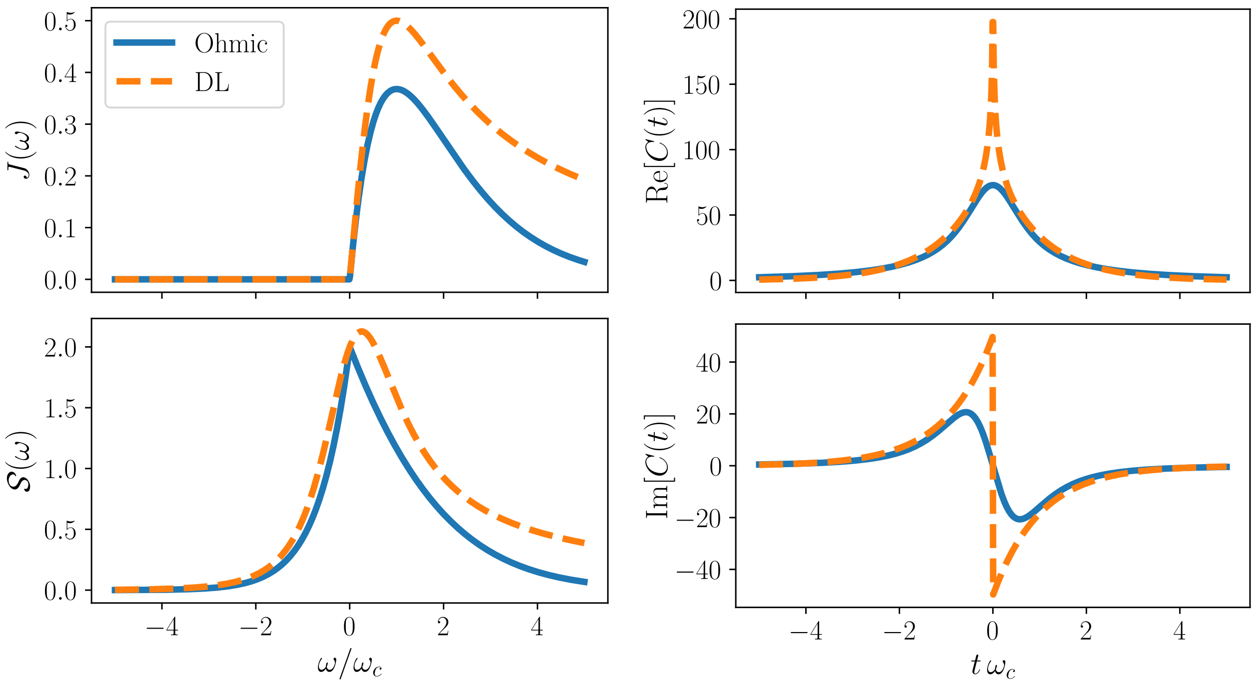

Figure 2.4: Comparison of representative bath models (Ohmic and Drude–Lorentz) showing spectral densities, associated power spectra, and time-domain correlation functions for coupling strength $\alpha = 0.1$, cutoff $\omega_c = 100$, and temperature $T = 100$. To make both models as comparable as possible, the Drude–Lorentz parameters are chosen consistently with the Ohmic-type convention via $\gamma = \omega_c$ and $\lambda = \alpha\,\omega_c/2$.

Interestingly, when the electronic transition frequency substantially exceeds the characteristic bath frequencies, the *high-frequency tails* of $J(\omega)$ become dominant for determining population relaxation kinetics and must be carefully specified, even if spectral densities with different functional forms appear nearly identical in their peak structures [33].

2.4.2 Structured Spectral Densities

Beyond these overdamped models, underdamped Brownian-oscillator spectral densities are often used to represent discrete vibrational structure, or microscopic chain models such as the Rubin model, whose continuum limits recover Ohmic/Drude-type behavior [30, 34]. In practice, structured environments are frequently approximated by a sum of such components, such as Drude plus Brownian terms, which is also a standard starting point for hierarchical-equations-of-motion (HEOM) approaches [35, 36].

A bath is *structured* when $J(\omega)$ exhibits pronounced features near such transitions, such as sharp resonances, band edges, or steep slopes [1]. By contrast to the Ohmic cases, a resonant Lorentzian peak centered at a finite frequency ω_0 may be written as

$$J(\omega) = \lambda \frac{\gamma}{(\omega - \omega_0)^2 + \gamma^2}, \tag{2.35}$$

where γ controls the linewidth and λ the coupling strength. Such a spectrum concentrates weight near ω_0 and corresponds physically to a damped mode (e.g. a molecular vibration or cavity-like resonance) coupled to a broader background bath. The associated bath correlation function contains oscillatory contributions of the form $e^{-\gamma t}\cos(\omega_0 t)$, reflecting the finite-frequency structure. The distinction between structured and unstructured baths is independent of Markovianity.

In summary, environmental effects enter the Redfield framework through $J(\omega)$, which determines the bath correlation function $C(t) = \langle E(t)E(0)\rangle$ and thus the memory time τ_E (Eqs. (2.31) to (2.32)). Broad, slowly varying spectra typically support a Markovian description, whereas sharply structured features near system transitions can generate oscillatory and long-lived correlations, requiring more careful modeling beyond simple Markovian Redfield theory.

Chapter 3

Nonlinear Spectroscopy and Two-Dimensional Electronic Spectra

3.1 Motivation and Historical Development

Spectroscopy studies how matter interacts with energy, most commonly through absorption, emission, or scattering of radiation [37]. Because atoms and molecules occupy discrete energy levels, optical transitions satisfy the Planck relation $\Delta E = h\nu = hc/\lambda$. Early spectroscopy exploited this to identify characteristic line spectra that served as fingerprints of quantized structure [38–40]. Quantum mechanics later connected these lines to transitions between eigenstates of the microscopic Hamiltonian [41, 42]. In practice, transitions are often labelled by the wavenumber $\tilde{\nu} = 1/\lambda$ [cm^{-1}], which is proportional to the transition energy. The electronic transitions studied in this thesis lie in the visible and near-UV range, $\tilde{\nu} \sim 13\,000$–$25\,000\,\mathrm{cm}^{-1}$, where coherence and relaxation dynamics unfold on femtosecond timescales.

For decades, spectroscopy was essentially linear: a weak field probes the sample and one records intensity as a function of frequency. Linear absorption and emission reveal excitation energies, oscillator strengths, and broadening mechanisms, such as natural linewidth or Doppler broadening [37]. However, a one-dimensional line shape compresses distinct dynamical processes—coherence loss, population transfer, and environment-induced frequency fluctuations—into a single spectral profile. This makes different microscopic mechanisms difficult to disentangle.

The induced polarisation $\boldsymbol{P}(t)$ contains not only components linear in the applied field $\boldsymbol{E}(t)$ [43],

$$\boldsymbol{P}^{(1)}(t) = \int_0^\infty \mathrm{d}\tau\, \boldsymbol{R}^{(1)}(\tau)\cdot\boldsymbol{E}(t-\tau),$$

where $\boldsymbol{R}^{(1)}$ is the causal linear-response kernel. There are also quadratic, cubic, and higher-order contributions,

$$\boldsymbol{P}^{(n)}(t) \propto \boldsymbol{E}(t)^n, \qquad n \geq 2.$$

These terms generate effects such as second-harmonic generation, four-wave mixing, and stimulated Raman scattering. Conceptually, they probe multi-time correlation functions of the system. Therefore, they encode dynamical information that is inaccessible in the linear regime [3, 44].

Two-dimensional Fourier-transform spectroscopy is a powerful tool within nonlinear spectroscopy. By correlating excitation and detection frequencies, it separates homogeneous from inhomogeneous broadening. Homogeneous broadening reflects the intrinsic linewidth of a single chromophore, set by coherence loss and population relaxation. Inhomogeneous broadening, by contrast, arises from static disorder in the molecular environment (e.g. different binding sites or solvent configurations), which distributes transition frequencies across the ensemble. This separation allows the homogeneous linewidth to be extracted even in strongly inhomogeneously broadened systems [46, 47]. In addition, two-dimensional spectra track spectral diffusion—time-dependent fluctuations of transition frequencies caused by a slowly evolving local environment—through the loss of frequency–frequency correlation with waiting time [3, 5]. Cross peaks further visualize couplings between transitions and energy-transfer pathways, which are central to the analyses in this thesis.

To connect these experimental capabilities to the theoretical framework developed in Ch. 2, this thesis uses the response-function formalism. It links a given pulse sequence to the measured signal via the induced polarisation. By separating matter and field contributions, this formalism makes the perturbative order explicit. The matter-side derivation—how incident fields generate $\boldsymbol{P}^{(n)}$—is given in App. B and shows how open-system master equations, such as the Redfield equation (microscopically derived in Sec. 2.1), enter spectroscopy. The field-side derivation—how $\boldsymbol{P}^{(3)}$ radiates the signal—is given in App. C.

The remainder of this chapter is organized as follows. Section 3.2 introduces the semiclassical light–matter Hamiltonian in the dipole approximation, which establishes the fundamental microscopic interaction governing how laser pulses perturb molecular systems and provides the starting point for all spectroscopic calculations in this thesis. Section 3.3 provides a unified summary of the two technical appendices by combining the matter-side response expansion with the Maxwell-side radiation step into one signal chain. Section 3.4 then states the core response-function expressions and pathway structure used in the spectroscopy workflow. Finally, Sec. 3.5–Sec. 3.7 applies these ingredients to three-pulse experiments, phase matching, phase cycling, and 2D spectrum construction.

3.2 Light–Matter Interaction

Throughout this thesis, the electromagnetic field is treated as a classical, time-dependent drive, while the matter degrees of freedom are described quantum mechanically. The central object is the density matrix of the matter, whose evolution determines the optical response.

The microscopic dynamics of the full system (molecule plus environment) are governed by the Liouville–von Neumann equation (cf. Equation (2.5))

$$\frac{d}{dt}\hat{\rho}_{\mathrm{T}}(t) = -i\,[\hat{H}(t), \hat{\rho}_{\mathrm{T}}(t)], \tag{3.1}$$

with total Hamiltonian

$$\hat{H}(t) = \hat{H}_{\mathrm{S}}(t) + \hat{H}_{\mathrm{E}} + \hat{H}_{\mathrm{int}}.$$

Here $\hat{H}_{\mathrm{E}}$ is the environment Hamiltonian and $\hat{H}_{\mathrm{int}}$ the system–environment coupling. The semiclassical light field enters only through the time-dependent system Hamiltonian $\hat{H}_{\mathrm{S}}(t)$. The structure therefore is a time-dependent version of the one used in Sec. 2.1.2.

3.2.1 Minimal Coupling

Mathematical details can be found in [41, 48]. In Coulomb gauge, the interaction of a particle of mass m and charge q with a classical electromagnetic field is described by the minimal-coupling Hamiltonian

$$\hat{H}(\hat{\boldsymbol{r}}, t) = \frac{1}{2m}\Big[\hat{\boldsymbol{p}} + q\,\boldsymbol{A}(\hat{\boldsymbol{r}}, t)\Big]^2 + V(\hat{\boldsymbol{r}}) - q\,\phi(\hat{\boldsymbol{r}}, t),$$

where $\boldsymbol{A}$ and ϕ are the vector and scalar potentials of the external field, V is the static binding potential, $\boldsymbol{p}$ is the canonical momentum operator, and $\hat{\boldsymbol{r}}$ is the position operator. The transverse electric field is related to the vector potential by

$$\boldsymbol{E}_{\perp}(\boldsymbol{r}, t) = -\frac{\partial \boldsymbol{A}(\boldsymbol{r}, t)}{\partial t}.$$

In most optical settings the wavelength is much larger than the molecular size, so the field varies negligibly across the system. In this limit, a unitary transformation leads to an equivalent effective Hamiltonian

$$\hat{H}_{\mathrm{S}}(t) = \hat{H}_0 - \hat{\boldsymbol{\mu}} \cdot \boldsymbol{E}(t),$$

with $\hat{H}_0$ the field-free Hamiltonian and $\hat{\boldsymbol{\mu}} = -q\,\hat{\boldsymbol{r}}$ the electric dipole operator. For a fixed linear polarisation direction $\boldsymbol{e}$, define

$$E(t) = \boldsymbol{e} \cdot \boldsymbol{E}(t), \qquad \hat{\mu} = \hat{\boldsymbol{\mu}} \cdot \boldsymbol{e},$$

so that

$$\hat{H}_{\mathrm{S}}(t) = \hat{H}_0 - \hat{\mu}\, E(t).$$

A key simplification used throughout is the rotating-wave approximation (RWA). It is introduced here and defined again in Eq. (3.26); its role in pathway selection is discussed in Sec. 3.5.3. For narrow-linewidth transitions driven near resonance, the field is decomposed into positive- and negative-frequency components $\boldsymbol{E}^{(+)}$ and $\boldsymbol{E}^{(-)}$, and only the near-resonant cross terms $\hat{\boldsymbol{\mu}}^{(+)}\cdot\boldsymbol{E}^{(-)}$ and $\hat{\boldsymbol{\mu}}^{(-)}\cdot\boldsymbol{E}^{(+)}$ remain in the equation of motion.

3.3 Microscopic and Macroscopic Response

This section summarizes, in one place, the two complementary derivations developed in the appendices. The next sections then build on this summary to detail the response-function formalism and its application to 2D spectroscopy.

For a representative molecule, the microscopic polarisation is the expectation value of the electric dipole operator,

$$\boldsymbol{P}(t) = \mathrm{Tr}_{\mathrm{S}}\big[\hat{\boldsymbol{\mu}}\,\hat{\rho}_{\mathrm{S}}(t)\big]. \tag{3.2}$$

Physically, $\boldsymbol{P}(t)$ quantifies the laser-driven redistribution of charge within the molecule. The field induces charge displacements, and these microscopic motions are the origin of the polarisation that ultimately acts as a source for the emitted electromagnetic signal. On a microscopic level, the polarisation density can be understood directly in terms of moving charges. Consider charges q_i at positions $\boldsymbol{r}_i(t)$:

$$\boldsymbol{P}(\boldsymbol{r}, t) \sim \sum_i q_i \boldsymbol{r}_i(t)\, \delta(\boldsymbol{r} - \boldsymbol{r}_i(t)).$$

The macroscopic polarisation field $\boldsymbol{P}(\boldsymbol{r}, t)$ is obtained by coarse-graining over a small volume. This volume must be large compared to molecular dimensions, but small on macroscopic scales. Appropriate averaging then yields the macroscopic quantity [62].

Here $\hat{\rho}_{\mathrm{S}}(t) = \mathrm{Tr}_{\mathrm{E}}\,\hat{\rho}_{\mathrm{T}}(t)$ is obtained by tracing out the environment from the total state, and it follows the reduced open-system dynamics introduced in Ch. 2. In practice, dissipative effects like population relaxation and dephasing enter through this reduced evolution, which provides the dynamical input to the matter-side response derivation in App. B.

In the perturbative regime, the macroscopic polarisation is expanded as

$$\boldsymbol{P} = \boldsymbol{P}^{(1)} + \boldsymbol{P}^{(3)} + \cdots .$$

The field-side derivation then yields the wave equation with polarisation source term,

$$\nabla \times \nabla \times \boldsymbol{E}(\boldsymbol{r}, t) + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{E}(\boldsymbol{r}, t) = -\mu_0 \frac{\partial^2}{\partial t^2}\boldsymbol{P}(\boldsymbol{r}, t).$$

In a four-wave-mixing experiment with heterodyne detection (see Sec. 3.7.1), the recorded quantity is linear in the emitted field and therefore in the selected third-order polarisation component,

$$\boldsymbol{E}_{\mathrm{S}}^{(3)}(\boldsymbol{r}, t) \propto i\, \boldsymbol{P}^{(3)}(\boldsymbol{r}, t).$$

This closes the link between microscopic dynamics and the measured signal: the material model determines $R^{(3)}$, $R^{(3)}$ determines $P^{(3)}$, and Maxwell propagation/detection map $P^{(3)}$ to the observable field. The combined result of both appendices is the practical spectroscopy chain used throughout this thesis: the third-order response function $R^{(3)}$ determines the third-order polarisation $P^{(3)}$, which generates the emitted signal field E_{S}, from which the detected signal $\tilde{E}_{\mathrm{S}}$ is obtained. Section 3.4 now details the first link, i.e. the response-function construction of $R^{(3)}$ and its pathway decomposition.

3.4 Response Function Formalism

The response-function formalism connects the applied field $\boldsymbol{E}(t)$ to the induced polarisation $\boldsymbol{P}^{(n)}(t)$. Here the key results are stated for linear response ($n = 1$) and, most importantly for this thesis, for third-order response ($n = 3$).

Linear response is the simplest setting in which the causal time-domain structure is explicit and the Liouville-space pathway structure first appears. For these reasons, it is discussed in detail.

3.4.1 Linear Response and the Kubo Formula

Specialising Eq. (B.15) and Eq. (B.16) to $n = 1$ gives

$$\boldsymbol{P}^{(1)}(t) = \int_0^{\infty} \mathrm{d}\tau_1\, \boldsymbol{R}^{(1)}(\tau_1)\, E(t - \tau_1), \tag{3.3}$$

with

$$\boldsymbol{R}^{(1)}(\tau_1) = \mathrm{i}\,\mathrm{Tr}\big\{\hat{\boldsymbol{\mu}}\,\mathcal{U}_0(\tau_1)\,\mathcal{V}[\hat{\rho}_{\mathrm{eq}}]\big\}. \tag{3.4}$$

Even at first order, Eq. (3.4) contains two Liouville-space pathways. Expanding the commutator gives a difference of two time-ordered dipole correlation functions, which motivates defining

$$J_1(\tau) \equiv \mathrm{Tr}\big\{\hat{\mu}_{\mathrm{I}}(\tau)\,\hat{\mu}_{\mathrm{I}}(0)\,\hat{\rho}_{\mathrm{eq}}\big\}, \tag{3.5}$$

so that the causal response can be written as

$$R^{(1)}(\tau) = i\,\Theta(\tau)\,\big[J_1(\tau) - J_1^*(\tau)\big]. \tag{3.6}$$

The two terms correspond to interactions on the ket and on the bra, with a relative minus sign.

To make this structure explicit, consider the simplest example of a two-level system. This allows the general expressions to be evaluated in closed form and introduces the double-sided Feynman representation that will later be used for higher-order processes (see Sec. 3.4.2).

Two-Level System

Consider a system with electronic states $|g\rangle$ and $|e\rangle$. For optical transitions ($\omega_{eg} \sim \tilde{\nu}_0 \approx 16\,000\,\mathrm{cm}^{-1}$), the excited-state thermal population at room temperature is negligible. Therefore, the initial condition can be approximated as the electronic ground state,

$$\hat{\rho}_{\mathrm{eq}} = |g\rangle\,\langle g|\,.$$

Write the dipole operator in terms of the transition dipole moment d and a dimensionless operator $\hat{m}$,

$$\hat{\mu} = d\,\hat{m} \quad \text{with} \quad \hat{m} = |e\rangle\,\langle g| + |g\rangle\,\langle e|\,.$$

The pathway logic is already visible here: a commutator produces a left and a right action with a relative minus sign. Expanding the first commutator converts the equilibrium state $\hat{\rho}_{\text{eq}}$ into a superposition of optical coherences,

$$\mathcal{V}[|g\rangle\langle g|] \approx |e\rangle\langle g| - |g\rangle\langle e| \,. \tag{3.7}$$

During the first delay τ, the coherences undergo free evolution. In the secular approximation, they evolve independently with the Green function

$$G_{eg}(\tau) = \exp\bigl[-(\mathrm{i}\omega_{eg} + \Gamma_{eg})\tau\bigr], \tag{3.8}$$

where $\omega_{eg} = E_e - E_g$ and Γ_{eg} is the optical dephasing rate. Applying the free-evolution propagator to each coherence then gives

$$\mathcal{U}_0(\tau)[|e\rangle\langle g|] \approx G_{eg}(\tau)\ |e\rangle\langle g| \,, \qquad \mathcal{U}_0(\tau)[|g\rangle\langle e|] \approx G^*_{eg}(\tau)\ |g\rangle\langle e| \,.$$

Double-Sided Feynman Diagrams

The left/right action distinction encoded in $\mathcal{V}[\hat{A}] \equiv [\hat{\mu}, \hat{A}]$ generates distinct pathways with characteristic sign patterns. These pathways are visualised by double-sided Feynman diagrams in Fig. 3.1. The construction rules are:

- Two vertical lines represent the ket (left) and bra (right) branches of $\hat{\rho}_{\text{S}}$; time flows upward.
- Both branches start in thermal equilibrium $|g\rangle$.
- Straight arrows mark interactions, i.e. actions of $\mathcal{V}$.
- Curly lines represent readout by $\hat{m}$, returning the system to $|g\rangle\,\langle g|$. Only closed diagrams contribute to the signal.
- Each right interaction contributes a factor -1, which fixes the overall sign.
- Coherences carry optical phase factors ($G_{eg}(\tau)$ for $|e\rangle\,\langle g|$, $G^*_{eg}(\tau)$ for $|g\rangle\,\langle e|$). Populations are approximately stationary.
- Under the RWA, each interaction is associated with $\boldsymbol{E}^{(+)}$ or $\boldsymbol{E}^{(-)}$, which determines the phase-matching direction (see Sec. 3.5.2).

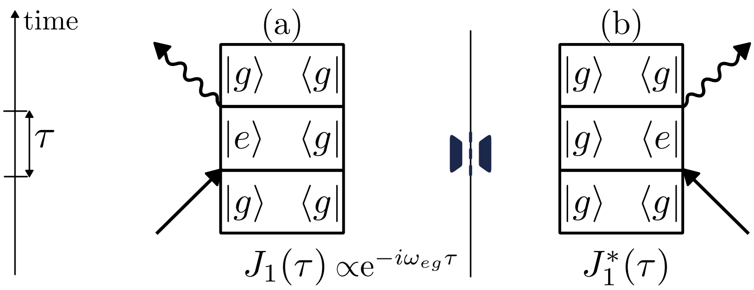

Figure 3.1: First-order double-sided Feynman diagrams contributing to linear response. (a) Interaction on the ket (left action). (b) Interaction on the bra (right action). The commutator in Eq. (3.4) is their difference [3].

Frequency-Domain Susceptibility and Absorption

The linear susceptibility is the Fourier transform of the causal response kernel Eq. (3.6),

$$\chi^{(1)}(\omega) = \int_{-\infty}^{\infty} \mathrm{d}\tau \, R^{(1)}(\tau) \, \mathrm{e}^{\mathrm{i}\omega\tau} = \int_{0}^{\infty} \mathrm{d}\tau \, R^{(1)}(\tau) \, \mathrm{e}^{\mathrm{i}\omega\tau}.$$

With this, Eq. (3.3) becomes in frequency space

$$\boldsymbol{P}^{(1)}(\omega) = \chi^{(1)}(\omega) \cdot \boldsymbol{E}(\omega).$$

Conventional line shapes can be connected to the pathway functions $J_1(\tau)$ and $J_1^*(\tau)$ defined in Eq. (3.5), by approximating the coherence pathway through the Green function, $J_1(\tau) \propto G_{eg}(\tau)$. This yields

$$\int_{0}^{\infty} \mathrm{d}\tau \, G_{eg}(\tau) \, \mathrm{e}^{\mathrm{i}\omega\tau} = \frac{1}{\Gamma_{eg} + \mathrm{i}(\omega_{eg} - \omega)},$$

and therefore the familiar complex Lorentzian susceptibility

$$\chi^{(1)}(\omega) \propto d^2 \frac{1}{\Gamma_{eg} - \mathrm{i}(\omega - \omega_{eg})}.$$

Its imaginary part gives the absorptive line shape centred at ω_{eg} with half-width Γ_{eg}, and the absorption coefficient satisfies $\alpha(\omega) \propto \operatorname{Im} \chi^{(1)}(\omega)$. For multiple excited states e_i, the total absorption is a sum over allowed $g \to e_i$ transitions.

3.4.2 Nonlinear Response

Even-order susceptibilities vanish in centrosymmetric media, so the lowest-order nonlinear response relevant here is third order [3, 49]. The central $\chi^{(3)}$ process for this thesis is four-wave mixing (FWM) [49], in which three incident fields generate a signal field. The resulting photon-echo and free-induction-decay contributions form the basis of 2D spectroscopy.

Third-Order Response Function $R^{(3)}$

Inserting $n = 3$ into Eq. (B.15) and Eq. (B.16) gives

$$\boldsymbol{P}^{(3)}(t) = \int_{0}^{\infty} \mathrm{d}\tau_3 \int_{0}^{\infty} \mathrm{d}\tau_2 \int_{0}^{\infty} \mathrm{d}\tau_1 \, R^{(3)}(\tau_3, \tau_2, \tau_1) \, E(t - \tau_3) \, E(t - \tau_3 - \tau_2) \, E(t - \tau_3 - \tau_2 - \tau_1), \quad (3.9)$$

with response function

$$R^{(3)}(\tau_3, \tau_2, \tau_1) = (\mathrm{i})^3 \, \mathrm{Tr}\Big\{ \hat{\boldsymbol{\mu}} \, \mathcal{U}_0(\tau_3) \, \mathcal{V} \, \mathcal{U}_0(\tau_2) \, \mathcal{V} \, \mathcal{U}_0(\tau_1) \, \mathcal{V}[\hat{\rho}_{\mathrm{eq}}] \Big\}. \quad (3.10)$$

This third-order response contains the multi-time correlations central to the present thesis. Reading the operator product in Eq. (3.10) from right to left corresponds to the causal sequence: start from $\hat{\rho}_{\mathrm{eq}}$, apply a dipole interaction, evolve freely, apply another interaction, and so on, followed by the dipole readout at time t. In a three-pulse experiment one writes $E(t) = E_1(t) + E_2(t) + E_3(t)$; inserting this into Eq. (3.9) generates many terms, and the experimentally relevant contribution (e.g. rephasing) is selected by phase matching or phase cycling. The material kernel responsible for that selection is precisely Eq. (3.10) [50].

Two-Level System and Pathway Expansion

To see concretely how higher-order expressions expand, consider again the two-level system from Sec. 3.4.1. After the first interaction $\mathcal{V}$, $\hat{\rho}_S$ is a linear combination of the coherences $|e\rangle\langle g|$ and $|g\rangle\langle e|$, followed by free evolution during τ_1, which imprints phase factors $G_{eg}(\tau_1)$ and $G^*_{eg}(\tau_1)$ (see Eqs. (3.7) and (3.8)). A second light–matter interaction converts these coherences into populations,

$$\mathcal{V}[|e\rangle\langle g|] \approx |g\rangle\langle g| - |e\rangle\langle e| \,, \qquad \mathcal{V}[|g\rangle\langle e|] \approx |e\rangle\langle e| - |g\rangle\langle g| \,.$$

During τ_2, the density operator is (in this simple model) a linear combination of populations. The third commutator converts populations back into coherences, which evolve during τ_3. Finally, left multiplication by $\hat{m}$ and the trace yield the measurable polarisation.

Writing the nested commutator Eq. (3.10) explicitly,

$$R^{(3)}(\tau_3,\tau_2,\tau_1) = (-i)\,\mathrm{Tr}\Big\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,[\hat{\mu}_\mathrm{I}(\tau_1+\tau_2),[\hat{\mu}_\mathrm{I}(\tau_1),[\hat{\mu}_\mathrm{I}(0),\hat{\rho}_\mathrm{eq}]]]\Big\},$$

one obtains the $2^3 = 8$ operator products

$$\begin{aligned}
R^{(3)}(\tau_3,\tau_2,\tau_1) = {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\,\hat{\mu}_\mathrm{I}(\tau_1)\,\hat{\mu}_\mathrm{I}(0)\,\hat{\rho}_\mathrm{eq}\} && \Rightarrow \quad R_4 && (3.11)\\
- {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\,\hat{\mu}_\mathrm{I}(\tau_1)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(0)\} && \Rightarrow \quad R_1^* && (3.12)\\
- {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\,\hat{\mu}_\mathrm{I}(0)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(\tau_1)\} && \Rightarrow \quad R_2^* && (3.13)\\
+ {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(0)\,\hat{\mu}_\mathrm{I}(\tau_1)\} && \Rightarrow \quad R_3 && (3.14)\\
- {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(\tau_1)\,\hat{\mu}_\mathrm{I}(0)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\} && \Rightarrow \quad R_3^* && (3.15)\\
+ {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(\tau_1)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(0)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\} && \Rightarrow \quad R_2 && (3.16)\\
+ {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_\mathrm{I}(0)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(\tau_1)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\} && \Rightarrow \quad R_1 && (3.17)\\
- {} & \mathrm{Tr}\{\hat{\mu}_\mathrm{I}(\tau_1+\tau_2+\tau_3)\,\hat{\rho}_\mathrm{eq}\,\hat{\mu}_\mathrm{I}(0)\,\hat{\mu}_\mathrm{I}(\tau_1)\,\hat{\mu}_\mathrm{I}(\tau_1+\tau_2)\} && \Rightarrow \quad R_4^*. && (3.18)
\end{aligned}$$

Along the same lines as in linear response, the terms labelled R^*_α are the complex conjugates of R_α for Hermitian $\hat{\mu}$ and a real equilibrium state. In typical 2DES calculations (RWA) one therefore groups the response as

$$R^{(3)}(\tau_3,\tau_2,\tau_1) = (-i)\sum_{\alpha=1}^{4}\Big[R_\alpha(\tau_3,\tau_2,\tau_1) - R^*_\alpha(\tau_3,\tau_2,\tau_1)\Big]. \tag{3.19}$$

The four pathway classes from Eq. (3.19) are shown in Fig. 3.2. For interpretation, the eight

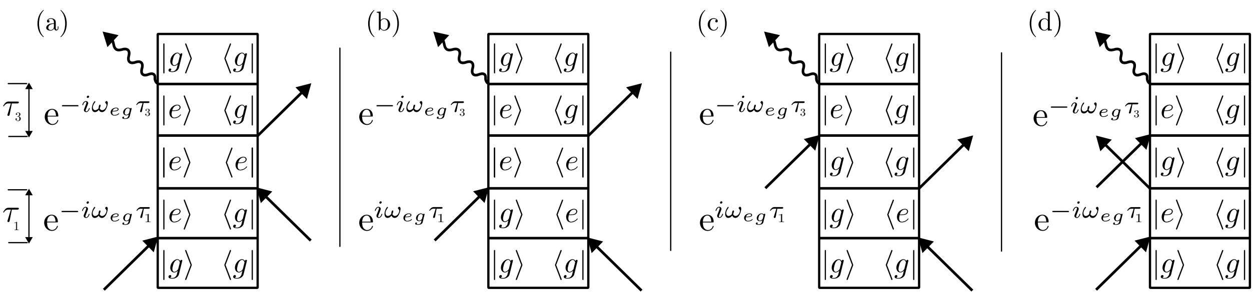

Figure 3.2: Double-sided Feynman diagrams for the four third-order pathway classes R_1–R_4 in Eqs. (3.11) to (3.18). Each diagonal arrow indicates a field interaction acting on the ket (left branch) or bra (right branch) of the density matrix. The complex-conjugate pathways R^*_α correspond to exchanging ket and bra actions as in Fig. 3.1.

third-order pathways are regrouped into physical classes [3, 5]:

- **Ground-state bleach (GSB).** During the waiting time τ_2 the system resides in the ground-state population $|g\rangle\langle g|$. The third interaction probes the depleted ground-state absorption.
- **Stimulated emission (SE).** During τ_2 the system resides in the excited-state population $|e\rangle\langle e|$. The third interaction stimulates emission back to the ground state.
- **Excited-state absorption (ESA).** During τ_2 the system is in $|e\rangle\langle e|$, but the third interaction promotes it to a higher-lying state $|f\rangle$. ESA is absent in the simple two-level as it requires an additional electronic state (see Sec. 4.3.5).

GSB and SE contribute with the same sign, whereas ESA enters with the opposite sign.

Phase factors of the individual pathways. Reading off the coherence Green function (cf. Section 3.4.2) for each Feynman diagram yields

$$\exp\left[-\mathrm{i}\omega_{eg}\, f_n(\tau_3, \tau_1)\right], \tag{3.20}$$

$$R_1\&R_4: \quad f_{1/4}(\tau_3, \tau_1) = \tau_3 + \tau_1, \tag{3.21}$$

$$R_2\&R_3: \quad f_{2/3}(\tau_3, \tau_1) = \tau_3 - \tau_1, \tag{3.22}$$

Populations during the second period τ_2 carry no rapid oscillation. However, during τ_2, population relaxation modifies the third-order pathways. For the two-level system, the dissipative channel can be described phenomenologically by the Lindblad jump operator $\hat{L} = \sqrt{\gamma_{ge}}\, |g\rangle\langle e|$, which implements $|e\rangle \to |g\rangle$ at rate γ_{ge}. The resulting equations of motion for the density-matrix elements are

$$\dot{\rho}_{ee} = -\gamma_{ge}\, \rho_{ee}, \qquad \dot{\rho}_{gg} = +\gamma_{ge}\, \rho_{ee}, \qquad \dot{\rho}_{eg} = -(\gamma_{ge}/2 + \gamma_{\varphi})\, \rho_{eg}, \tag{3.23}$$

where γ_{φ} accounts for additional pure dephasing. Solving these gives the population propagators

$$U_{ee,ee}(\tau_2) = \mathrm{e}^{-\gamma_{ge}\tau_2}, \qquad U_{gg,ee}(\tau_2) = 1 - \mathrm{e}^{-\gamma_{ge}\tau_2}, \qquad U_{gg,gg}(\tau_2) = 1, \tag{3.24}$$

and the total dephasing rate of the optical coherence is $\Gamma_{eg} = \gamma_{ge}/2 + \gamma_{\varphi}$, with $T_1 = 1/\gamma_{ge}$ and $T_2 = 1/\Gamma_{eg}$.

Consequently, every pathway through ρ_{ee} during τ_2 splits into two contributions: one remaining in ee (multiplied by $U_{ee,ee}$) and one relaxed to gg (multiplied by $U_{gg,ee}$).

For the two-level system, the third-order response $R^{(3)}$ thus contains contributions from GSB and SE only. In practice, $R^{(3)}$ is computed by summing the available pathways—contributions automatically generated by the Liouvillian propagator. Having established the third-order response function $R^{(3)}(\tau_3, \tau_2, \tau_1)$ in Sec. 3.4.2 as the material kernel encoding the microscopic dynamics, the experimental realisation is now specified: a sequence of three controllable laser pulses. This enables the isolation of different subsets of pathways via phase matching or phase cycling, in particular separating rephasing and nonrephasing signals.

3.5 Three-Pulse Experiments and Phase Matching

In a three-pulse experiment, the measured third-order signal is obtained by driving the system with a sequence of fields and reading out the radiated polarisation. For finite-duration pulses with partial overlap, the separation into strictly time-ordered interactions is approximate and the signal becomes a convolution of the material response with the pulse envelopes, which modifies pathway weights and phases [50, 51]. In this thesis these effects are captured directly by the nonperturbative simulations.

3.5.1 Electric Fields

The applied field is written as a sum of three pulses with carrier wavevectors $\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3$ and carrier frequency ω_L,

$$\boldsymbol{E}(\boldsymbol{r},t) = \sum_{j=1}^{3} \Big[\boldsymbol{E}_j(t-t_j)\, e^{\mathrm{i}(\boldsymbol{k}_j\cdot\boldsymbol{r}-\omega_L t+\phi_j)} + \text{c.c.}\Big], \tag{3.25}$$

where t_j denotes the arrival time of pulse j, ϕ_j its carrier phase, and $\boldsymbol{E}_j(t)$ a complex envelope. In the rotating-wave approximation, the field is decomposed into positive- and negative-frequency parts,

$$\boldsymbol{E}(\boldsymbol{r},t) = \boldsymbol{E}^{(+)}(\boldsymbol{r},t) + \boldsymbol{E}^{(-)}(\boldsymbol{r},t), \qquad \boldsymbol{E}^{(-)} = \big[\boldsymbol{E}^{(+)}\big]^*, \tag{3.26}$$

with analytic part

$$\boldsymbol{E}^{(+)}(\boldsymbol{r},t) = \sum_{j=1}^{3} \boldsymbol{E}_j(t-t_j)\, e^{\mathrm{i}(\boldsymbol{k}_j\cdot\boldsymbol{r}-\omega_L t+\phi_j)}.$$

Under the RWA, only the near-resonant cross terms $\hat{\boldsymbol{\mu}}^{(+)}\cdot\boldsymbol{E}^{(-)}$ and $\hat{\boldsymbol{\mu}}^{(-)}\cdot\boldsymbol{E}^{(+)}$ are retained [3, 49]. Consequently, contributions to the nonlinear polarisation acquire phase factors $\pm\boldsymbol{k}_j$ and $\pm\phi_j$. These are exploited for pathway selection by phase matching and phase cycling (see Sec. 3.6). An optional Bloch-sphere intuition for carrier-phase kicks and pulse area is provided in App. D; the main text here continues with the phase-matching formulation used in the simulations.

3.5.2 Wavevector Phase-Matching Condition

In nonlinear wave mixing, the signal is emitted into directions that satisfy energy and momentum conservation, $\Delta\boldsymbol{k} \approx 0$. In a plane-wave picture, momentum conservation implies that the signal wavevector $\boldsymbol{k}_\mathrm{S}$ is a signed sum of the input wavevectors,

$$\boldsymbol{k}_\mathrm{S} = \pm\boldsymbol{k}_1 \pm \boldsymbol{k}_2 \pm \boldsymbol{k}_3. \tag{3.27}$$

This is the *phase-matching condition*. The signs track whether the interaction involves $\boldsymbol{E}^{(+)}$ or $\boldsymbol{E}^{(-)}$ (see Eq. (3.26)). Equivalently, in Liouville space they track whether the interaction acts on the ket or bra branch of the density matrix: absorption-like interactions contribute $+\boldsymbol{k}_j$, while emission-like/complex-conjugate interactions contribute $-\boldsymbol{k}_j$.

Different sign choices correspond to different phase-matched signal directions and hence to different subsets of Liouville-space pathways. In noncollinear geometries these directions can be spatially separated. The experimental geometry considered here is shown in Fig. 3.3; it defines $\boldsymbol{k}_1$, $\boldsymbol{k}_2$, $\boldsymbol{k}_3$ and the signal direction $\boldsymbol{k}_\mathrm{S}$ used below.

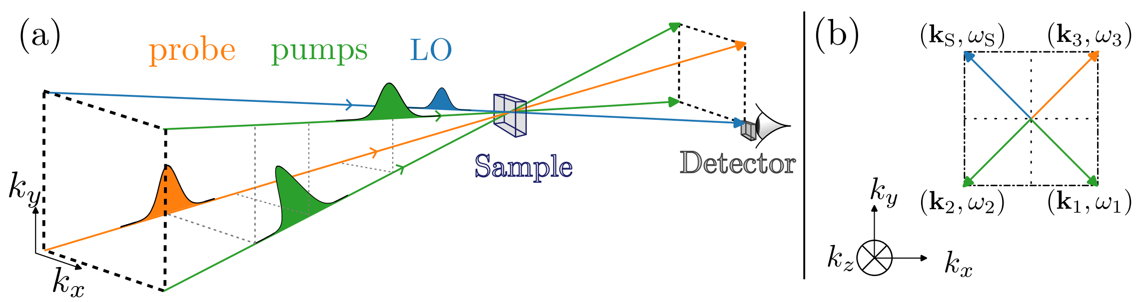

Figure 3.3: Noncollinear BOXCARS geometry. (a) Beam arrangement in the k_x–k_y plane with three input beams (green/orange) and the signal direction $\boldsymbol{k}_\mathrm{S}$ (blue); shaded envelopes indicate relative pulse timing. (b) Wavevector diagram showing the relation between $\boldsymbol{k}_1$, $\boldsymbol{k}_2$, $\boldsymbol{k}_3$, and $\boldsymbol{k}_\mathrm{S}$, with the corresponding frequency labels ω_j and ω_S.

3.5.3 Rephasing and Nonrephasing Phase-Matching Conditions

Inserting the three-pulse field Eq. (3.25) into the third-order convolution Eq. (3.9) and allowing only RWA-resonant terms, the third-order polarisation decomposes into components labelled by a sign triple (l, m, n) with $l, m, n \in \{-1, +1\}$,

$$\boldsymbol{P}^{(3)}(\boldsymbol{r}, t) = \sum_{l,m,n} e^{\mathrm{i}(\boldsymbol{k}_\mathrm{S}\cdot\boldsymbol{r} - \omega_L t + \Phi_{lmn})}\, \boldsymbol{P}^{(3)}_{l,m,n}(t), \qquad \boldsymbol{k}_\mathrm{S} = l\boldsymbol{k}_1 + m\boldsymbol{k}_2 + n\boldsymbol{k}_3, \quad \Phi_{lmn} = l\phi_1 + m\phi_2 + n\phi_3. \tag{3.28}$$

The triple (l, m, n) labels the sign choices for each of the three interactions; as noted above, these determine the ket/bra assignment for each interaction step via the phase-matching condition. Only combinations satisfying the macroscopic phase-matching condition contribute efficiently in an extended sample [49].

The two third-order directions of interest for 2DES are

$$\boldsymbol{k}_\mathrm{R} = -\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3, \tag{3.29}$$

$$\boldsymbol{k}_\mathrm{NR} = +\boldsymbol{k}_1 - \boldsymbol{k}_2 + \boldsymbol{k}_3, \tag{3.30}$$

usually identified as *rephasing* (R) and *nonrephasing* (NR) directions, respectively [3, 5, 47].

Besides the RWA, a useful limiting case is the impulsive limit of well separated pulses, where the interaction times are strictly ordered and pulse overlap can be neglected [3, 5]. Then the sign patterns in Eqs. (3.29) to (3.30) can be read directly as a sequence of ket- and bra-branch interactions (see App. D.1).

For perfectly ordered interactions, pulses 1 and 3 determine whether the coherence phase evolves with opposite signs or with the same sign during τ_1 and τ_3. Opposite signs produce the rephasing contribution; equal signs produce the nonrephasing contribution.

For the rephasing signal, the sign reversal acts like a photon echo and refocuses static disorder in an inhomogeneously broadened ensemble [4, 5, 47]. The nonrephasing signal does not refocus static disorder. However it gives a free-induction–decay response and is therefore directly sensitive to homogeneous linewidths [4]. Consistently, from Eq. (3.22), R_2 and R_3 carry $\exp[-\mathrm{i}\omega_{eg}(\tau_3 - \tau_1)]$ (rephasing), whereas R_1 and R_4 carry $\exp[-\mathrm{i}\omega_{eg}(\tau_3 + \tau_1)]$ (nonrephasing).

As a note, other third-order pathways like *double-quantum coherence* can be accessed by the phase matching condition $\boldsymbol{k}_\mathrm{S} = +\boldsymbol{k}_1 + \boldsymbol{k}_2 - \boldsymbol{k}_3$. They are not required for the aims of this thesis. See Ref. [53] for a broader overview.

3.6 Phase Cycling and Selection of Pathways

In a collinear geometry, different signal wavevector components $\boldsymbol{k}_\mathrm{S}$ cannot be spatially separated.[1] Instead, pathway selection is achieved by *phase cycling*: one repeats the measurement for different carrier-phase settings ϕ_j and isolates the desired contribution by a discrete Fourier projection with respect to (ϕ_1, ϕ_2, ϕ_3) [3, 54]. This works because each Liouville-space pathway acquires a characteristic phase factor determined by the sequence of light–matter interactions.

The pulse sequence and delay definitions used throughout this work are shown in Fig. 3.4.

The third-order polarisation is 2π-periodic in each ϕ_j and can be expanded as a Fourier series,

$$P^{(3)}(t; \phi_1, \phi_2, \phi_3) = \sum_{l,m,n} e^{\mathrm{i}(l\phi_1 + m\phi_2 + n\phi_3)}\, P^{(3)}_{l,m,n}(t),$$

[1] In numerical simulations this limitation is even more direct: spatial propagation and beam geometry are typically not modelled, so $\boldsymbol{k}$-selection cannot be implemented at the level of the fields.

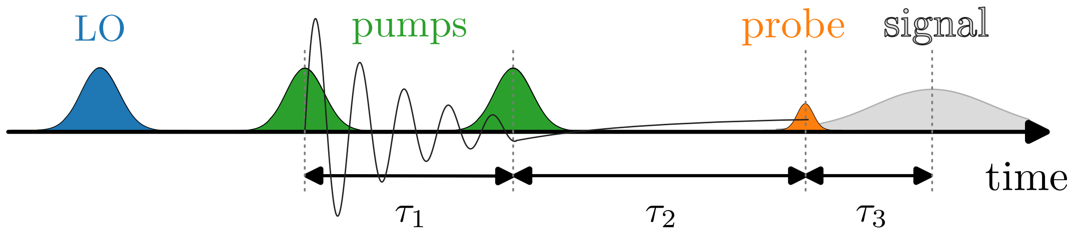

Figure 3.4: Collinear pulse sequence used for phase cycling. The local oscillator (LO, see Sec. 3.7.1), two pump pulses, the probe pulse, and the emitted signal are shown schematically in time. The three inter-pulse delays are labelled τ_1, τ_2, and τ_3, with the corresponding time variables t_{coh}, t_{wait}, t_{det} indicated below.

where $P^{(3)}_{l,m,n}(t)$ collects all third-order contributions carrying the phase combination (l, m, n). In particular, the rephasing and nonrephasing components correspond to

$$\text{rephasing:} \quad (l, m, n) = (-1, +1, +1), \tag{3.31}$$

$$\text{nonrephasing:} \quad (l, m, n) = (+1, -1, +1). \tag{3.32}$$

These index triples are the phase analogues of the wavevector combinations Eqs. (3.29) and (3.30) [3, 5, 54].

In practice, one evaluates $P^{(3)}$ on a finite phase grid, typically $\phi_j \in \{0, 2\pi/N_\phi, \ldots, 2\pi(N_\phi - 1)/N_\phi\}$ for each j (often $N_\phi = 4$, i.e. steps of $\pi/2$), which is sufficient to separate the third-order components relevant for rephasing and nonrephasing 2D spectra. The desired component is then obtained by discrete Fourier projection,

$$P^{(3)}_{l,m,n}(t) \simeq \frac{1}{N_\phi^3} \sum_{\phi_1,\phi_2,\phi_3} e^{-\mathrm{i}(l\phi_1 + m\phi_2 + n\phi_3)}\, P^{(3)}(t; \phi_1, \phi_2, \phi_3). \tag{3.33}$$

In the continuous limit this becomes the inverse Fourier transform

$$P^{(3)}_{l,m,n}(t) = \frac{1}{(2\pi)^3} \int_0^{2\pi}\!\int_0^{2\pi}\!\int_0^{2\pi} d\phi_1\, d\phi_2\, d\phi_3\; e^{-\mathrm{i}(l\phi_1 + m\phi_2 + n\phi_3)}\, P^{(3)}(t; \phi_1, \phi_2, \phi_3). \tag{3.34}$$

In the simulations used in this thesis, $P^{(3)}(t; \phi_1, \phi_2, \phi_3)$ is obtained by solving the full master equation, including the time-dependent light–matter interaction, for each phase setting (ϕ_1, ϕ_2, ϕ_3). The phase-selected third-order contribution is then extracted via Eq. (3.33), choosing (l, m, n) according to Eqs. (3.31) and (3.32).

3.7 Construction of Two-Dimensional Spectra

In a three-pulse experiment it is customary to parameterise the signal by the three time delays

$$t_{\mathrm{coh}} \equiv \tau_1, \qquad t_{\mathrm{wait}} \equiv \tau_2, \qquad t_{\mathrm{det}} \equiv \tau_3,$$

with delay variables defined in Eq. (B.10) and used in the third-order convolution Eq. (B.15). The quantities t_{coh}, t_{wait}, and t_{det} are referred to as coherence time, waiting time, and detection time, respectively.

Scanning the waiting time t_{wait} probes population dynamics and spectral diffusion during the population interval between the second and third interactions [3, 5].

The phase-cycled complex signal field $E_{\mathrm{R/NR}}(t_{\mathrm{coh}}, t_{\mathrm{wait}}, t_{\mathrm{det}})$ is the object from which 2D spectra are constructed. The Maxwell-side propagation and phase-matching assumptions underpinning

$E_S \propto P^{(3)}$ are derived in App. C. Together with the third-order material response from Eqs. (3.9) and (3.10), this gives the central chain used throughout the thesis,

$$R^{(3)} \implies P^{(3)} \implies E_S \implies \tilde{E}_S. \tag{3.35}$$

To make explicit how the recorded quantity connects to this chain, the next subsection discusses the detection step.

3.7.1 Detection: From Polarisation to Signal Field

Detectors measure intensities. In 2DES, the outgoing field is typically spectrally dispersed and recorded as a function of detection frequency ω_{det}. Formally, the recorded signal is proportional to the detected intensity integrated over the detector response time,

$$\tilde{E}_S(\omega_{\text{det}}) \propto \int I_{\text{det}}(\omega_{\text{det}}, t)\,dt. \tag{3.36}$$

Two common schemes differ in sensitivity and in whether phase information is retained: homodyne and heterodyne detection [55].

Homodyne (self-heterodyne) detection. In *homodyne* detection one records the intensity of the signal field itself,

$$I_{\text{HOD}}(\omega_{\text{det}}) \propto \left|E_S(\omega_{\text{det}})\right|^2. \tag{3.37}$$

This scales quadratically with the signal field (and thus with the underlying polarisation), mixes different nonlinear orders, and does not directly retain the signal phase. Many linear experiments can be viewed as a special "self-heterodyne" limit where the weak generated field interferes with a strong transmitted reference field that is automatically present.

Heterodyne detection (explicit LO). In background-free four-wave-mixing geometries the signal is emitted into a distinct phase-matched direction, so no strong reference beam co-propagates with it. One therefore introduces an explicit *local oscillator* (LO) field E_{LO} that overlaps with the signal on the detector. Writing the spectrally resolved total field as

$$E_{\text{tot}}(\omega_{\text{det}}) = E_{\text{LO}}(\omega_{\text{det}}) + E_S(\omega_{\text{det}}), \tag{3.38}$$

the measured spectral intensity is

$$I_{\text{HET}}(\omega_{\text{det}}) \propto \left|E_{\text{tot}}(\omega_{\text{det}})\right|^2 = |E_{\text{LO}}(\omega_{\text{det}})|^2 + |E_S(\omega_{\text{det}})|^2 + 2\,\text{Re}\left[E^*_{\text{LO}}(\omega_{\text{det}})E_S(\omega_{\text{det}})\right]. \tag{3.39}$$

For $|E_S| \ll |E_{\text{LO}}|$ the $|E_S|^2$ term can be neglected, and the LO background $|E_{\text{LO}}|^2$ can be measured and subtracted. It is convenient to define a normalised heterodyne signal,

$$\tilde{E}_{\text{HET}}(\omega_{\text{det}}) \equiv \frac{I_{\text{HET}}(\omega_{\text{det}}) - |E_{\text{LO}}(\omega_{\text{det}})|^2}{|E_{\text{LO}}(\omega_{\text{det}})|^2} \approx 2\,\text{Re}\left[\frac{E^*_{\text{LO}}(\omega_{\text{det}})E_S(\omega_{\text{det}})}{|E_{\text{LO}}(\omega_{\text{det}})|^2}\right], \tag{3.40}$$

which makes explicit that heterodyne detection renders the measurement linear in the signal field.

Combining this with the field–polarisation relation from App. C and the third-order polarisation from Eqs. (3.9) and (3.10) yields the proportionality used in this thesis,

$$\tilde{E}_{\text{HET}} \propto \text{Re}[E^*_{\text{LO}}E_S] \propto \text{Re}\left[E^*_{\text{LO}}P^{(3)}\right] \propto \text{Re}\left[E^*_{\text{LO}}\left(R^{(3)} * E_1 * E_2 * E_3\right)\right]. \tag{3.41}$$

Thus, once the phase-matched/phase-cycled third-order component is isolated (Sec. 3.5.2 and Sec. 3.6), the detected heterodyne signal reports the third-order response convolved with the applied pulse sequence.

Phase sensitivity and phasing. Writing the LO as $E_{\text{LO}}(\omega_{\text{det}}) = A(\omega_{\text{det}})e^{i\phi_{\text{LO}}(\omega_{\text{det}})}$ shows that the recorded cross term depends on the (in general unknown) LO spectral phase,

$$\tilde{E}_{\text{HET}}(\omega_{\text{det}}) \propto \text{Re}\left\{e^{-i\phi_{\text{LO}}(\omega_{\text{det}})}\, E_{\text{S}}(\omega_{\text{det}})\right\}. \tag{3.42}$$

If $\phi_{\text{LO}}(\omega_{\text{det}})$ is not controlled or calibrated, absorptive and dispersive contributions are mixed. Operationally, one reconstructs a complex signal (e.g. by spectral interferometry) and then fixes the remaining overall phase by a standard *phasing* procedure (often via a pump–probe projection) [**hybletal1998twodimensionalelectronicspectroscopy**]. After this calibration, the reconstructed complex E_{S} can be compared directly to theory.

In the numerical workflow of this thesis, the complex phase-selected signal is obtained directly from phase cycling (Eqs. (3.33) and (3.34)), so the unknown experimental LO phase does not appear as an additional fitting parameter.

Remark (what is actually measured, and how to recover a complex signal). Even with heterodyne detection, the detector output at fixed waiting time T and detection frequency is a *real* function of the coherence time. In an idealised description one may write the measured (rephasing) trace as

$$\tilde{E}_{\text{R}}^{(\text{meas})}(\omega_{\text{det}}; T, t_{\text{coh}}) = \text{Re}\left[e^{i\phi}\, \tilde{E}_{\text{R}}(\omega_{\text{det}}; T, t_{\text{coh}})\right], \tag{3.43}$$

where ϕ is an (unknown) global phase set by the relative signal/LO optical paths. This is the global phase ambiguity discussed above in frequency-resolved form (cf. Equation (3.42)). In experiments one reconstructs a complex signal and removes this by interferometric calibration plus phasing. In this thesis, complex $\tilde{E}_{\text{R/NR}}(t_{\text{coh}}, T, t_{\text{det}})$ is obtained directly from phase cycling in the simulations, and the extracted complex signal is then used for the Fourier-domain construction described in Sec. 3.7.

Remark (positive vs. negative coherence time). In some presentations one also considers negative t_{coh} values, which correspond to reversing the order of pulses 1 and 2. Then the rephasing photon-echo picture does not apply. Instead one obtains a nonrephasing free-induction–decay-like contribution complementary to the $t_{\text{coh}} > 0$ rephasing branch [56]. For each fixed waiting time t_{wait}, the signal is a function of the two coherence times t_{coh} and t_{det},

$$E_{\text{R/NR}} = E_{\text{R/NR}}(t_{\text{coh}}, t_{\text{wait}}, t_{\text{det}}).$$

3.7.2 Double Fourier Transform and 2D Spectra

After computing the time-domain polarisation (Eq. (3.2)) and selecting the desired phase-tagged component via phase cycling (Eq. (3.34)), one obtains the corresponding complex signal field $E_{\text{R/NR}}(t_{\text{coh}}, t_{\text{wait}}, t_{\text{det}})$ (Eq. (C.34)). For the nonperturbative numerical extraction, see Eq. (A.1) in App. A. The two-dimensional spectrum is defined as the double Fourier transform of the time-domain signal with respect to $(t_{\text{coh}}, t_{\text{det}})$ at fixed t_{wait}. The opposite sign in the t_{coh} exponent distinguishes rephasing (photon-echo) and nonrephasing branches [5, 54].

In the numerical implementation, slowly varying complex envelopes are used (the optical carrier $e^{-i\omega_L t}$ is removed), so ω_{coh} and ω_{det} represent Fourier variables of these envelopes.

For the rephasing signal,

$$\tilde{E}_{\text{R}}(\omega_{\text{coh}}, t_{\text{wait}}, \omega_{\text{det}}) = \int_0^\infty dt_{\text{det}} \int_0^\infty dt_{\text{coh}}\, e^{+i\omega_{\text{coh}}t_{\text{coh}}}\, e^{-i\omega_{\text{det}}t_{\text{det}}}\, E_{\text{R}}(t_{\text{coh}}, t_{\text{wait}}, t_{\text{det}}),$$

while for the nonrephasing signal,

$$\tilde{E}_{\text{NR}}(\omega_{\text{coh}}, t_{\text{wait}}, \omega_{\text{det}}) = \int_0^\infty dt_{\text{det}} \int_0^\infty dt_{\text{coh}}\, e^{-i\omega_{\text{coh}}t_{\text{coh}}}\, e^{-i\omega_{\text{det}}t_{\text{det}}}\, E_{\text{NR}}(t_{\text{coh}}, t_{\text{wait}}, t_{\text{det}}).$$

In numerical implementations, finite integration limits (discrete time grids) are used and discrete Fourier transforms are applied with appropriate windowing. On uniform grids these are evaluated efficiently via (shifted) fast Fourier transforms [5, 54].

The absorptive 2D spectrum is obtained as the real part of the sum

$$\tilde{E}_{\mathrm{abs}}(\omega_{\mathrm{coh}}, t_{\mathrm{wait}}, \omega_{\mathrm{det}}) = \mathrm{Re}\left\{\tilde{E}_{\mathrm{R}}(\omega_{\mathrm{coh}}, t_{\mathrm{wait}}, \omega_{\mathrm{det}}) + \tilde{E}_{\mathrm{NR}}(\omega_{\mathrm{coh}}, t_{\mathrm{wait}}, \omega_{\mathrm{det}})\right\}.$$

This combination removes dispersive distortions that appear when considering $|\tilde{E}_{\mathrm{R}}|$ or $|\tilde{E}_{\mathrm{NR}}|$ alone and yields peak shapes directly comparable with experimental absorptive 2D spectra [4, 57]. The construction outlined in this chapter—from the master equation through phase-cycled polarisations to the absorptive 2D spectrum—forms the computational backbone of the numerical results presented in the following chapters.

Chapter 4

Model Systems

Having established the spectroscopy workflow in Ch. 3 and the open-system dynamics tools in Ch. 2, the concrete model Hamiltonians and dipole operators used in the numerical simulations are now specified. The discussion proceeds from minimal reference systems to more structured ensembles:

- a **qubit** as the simplest benchmark for rephasing/nonrephasing signals.
- a **dimer** (two coupled sites) with a coherent coupling constant J.
- an N**-site ensemble** with a prescribed geometry, simulated in the *ground*, *single-excitation*, and *double-excitation* manifolds $|g\rangle, \{|n\rangle\}, \{|f\rangle\}$ as introduced in Ch. 3.

Throughout, standard third-order spectroscopy concepts [3–5] and the master-equation propagation methods from Ch. 2 are used.

4.1 Qubit (Reference Model)

The simplest nontrivial benchmark is a qubit, i.e. a two-level quantum system with ground state $|g\rangle$ and excited state $|e\rangle$.

Although a strict two-level model cannot capture excited-state absorption into a separate higher manifold, it is a clean test case for the basic workflow (pulse sequence $\to$ induced polarisation $\to$ rephasing/nonrephasing signals and 2D spectra) and for disentangling homogeneous dephasing, population relaxation, and inhomogeneous broadening.

4.1.1 Hamiltonian and Dipole Operator

The field-free system Hamiltonian is taken as

$$\hat{H}_{\mathrm{S}}^{(\mathrm{TLS})} = \omega_{eg} \, |e\rangle\langle e| \, , \tag{4.1}$$

with transition frequency ω_{eg}. The optical dipole operator (projected onto the chosen polarisation direction) is written as

$$\hat{\mu}_{\mathrm{S}}^{(\mathrm{TLS})} = \mu_{ge} \, |g\rangle\langle e| + \mu_{eg} \, |e\rangle\langle g| \, , \qquad \mu_{eg} = \mu_{ge}^{*}. \tag{4.2}$$

4.1.2 System–Bath Coupling Types

For a two-level system it is instructive to distinguish two types of system–bath coupling: *longitudinal* (or *pure-dephasing*) couplings are proportional to σ_z and commute with the bare qubit Hamiltonian in Eq. (4.1), so they do not induce transitions between energy eigenstates but instead cause the energy splitting to fluctuate in time, leading to decay of coherences without population relaxation (T_ϕ). In contrast, *transverse* couplings (e.g., σ_x, σ_y) can induce transitions and drive population relaxation (T_1).

4.1.3 Open-System Dynamics and Broadening Scenarios

The reduced dynamics are generated by the open-system Liouvillian introduced in Ch. 2. In practice, either a Redfield-type generator or (when appropriate) a Lindblad form is used.

The qubit is examined under two broadening scenarios:

- **Homogeneous broadening**: a single transition frequency ω_{eg} with dephasing/relaxation entirely due to the bath.
- **Inhomogeneous broadening**: an ensemble of realizations with slightly different ω_{eg} values (static disorder), and spectra obtained by averaging the resulting signals as discussed in Ch. 3.

4.1.4 Results: Rephasing Signal and Qubit 2D Spectra

To illustrate the meaning of rephasing and the difference between homogeneous and inhomogeneous broadening, (i) a representative time-domain rephasing trace and (ii) sequences of TLS 2D spectra at different waiting times T are included. The figures are inserted below as placeholders. If the image files are not present yet, compilation will fall back to a framed box.

To validate the numerical workflow, a simpler phenomenological Lindblad-type evolution is first reproduced using the standard RWA equations of motion for the density-matrix elements. With the ansatz $\rho_{eg} = \sigma_{eg} e^{-i\Omega t}$, the resulting equations are

$$\dot{\sigma}_{eg} = -\big(\Gamma + i\Delta\big)\, \sigma_{eg} + i\, \mu_{eg} E(t)\, (\rho_{gg} - \rho_{ee}), \tag{4.3}$$

$$\dot{\rho}_{ee} = -\gamma\, \rho_{ee} + i\, \mu_{eg} E(t)\, \sigma_{ge} - i\, \mu_{eg} E^*(t)\, \sigma_{eg}, \tag{4.4}$$

$$\dot{\rho}_{gg} = -i\, \mu_{eg} E(t)\, \sigma_{ge} + i\, \mu_{eg} E^*(t)\, \sigma_{eg}. \tag{4.5}$$

with detuning $\Delta = \omega_{eg} - \Omega$. This is for a resonant coupling to the field ($\Delta = 0$) exactly the optical Bloch equations as presented in [58]. It uses phenomenological decay rates Γ and γ for the coherence and population, respectively. By choosing $\Gamma = \gamma/2 + 1/T_2^*$, the pure-dephasing contribution T_2^* can be separated from the population-relaxation contribution $\gamma/2$ to the total dephasing rate. In the reproduction shown in Fig. 4.1, the parameters match the paper and the polarisation is sampled every 2 fs for τ and t in $[0, 600]$ fs. The field intensity is chosen so that the excited-state population remains below 1% ($E_0 = 0.05$), ensuring higher-order nonlinearities remain negligible.

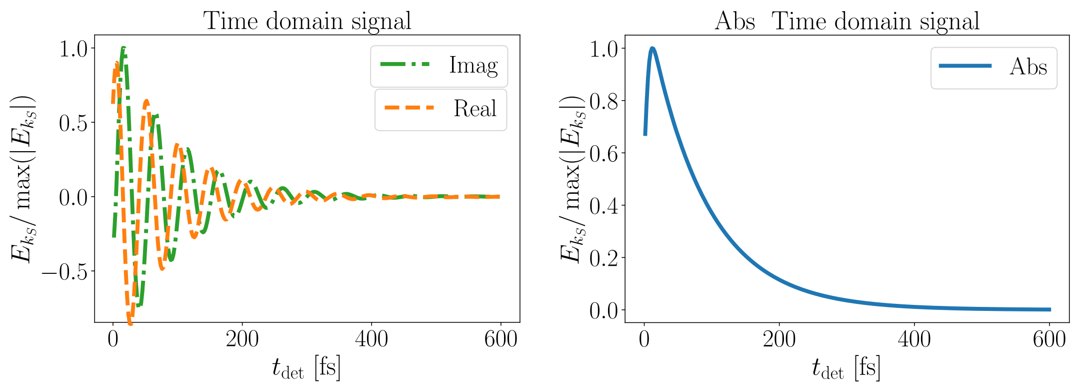

Figure 4.1: Time-domain three-pulse photon-echo (rephasing) signal of a two-level system computed from the RWA equations of motion, Eq. (4.3)–Eq. (4.5), with the paper parameters ($\tau = 300$ fs, $T = 1000$ fs, $\gamma^{-1} = 300$ fs, $T_2^* = 100$ fs, $\tau_{\mathrm{pulse}} = 15$ fs, $\omega_{eg} = 16000\,\mathrm{cm}^{-1}$, and inhomogeneous FWHM $\Delta = 200\,\mathrm{cm}^{-1}$) at resonance $\omega_L = \omega_{eg}$. This reproduces the reference result of [58].

The homogeneous case is shown in Fig. 4.2.

The inhomogeneous case is shown in Fig. 4.3.

Placeholder: qubit 2D spectra vs waiting time T (homogeneous broadening).
Put the file at `latex/figures/tls_2d_spectra_homogeneous.pdf`.

Figure 4.2: Sequence of rephasing/nonrephasing (or absorptive) 2D spectra of a homogeneously broadened qubit for increasing waiting time T. Solid/dashed contours indicate positive/negative contributions as in standard 2D-FTS conventions. (This placeholder should be replaced by the corresponding figure set.)

Placeholder: qubit 2D spectra vs waiting time T (inhomogeneous broadening).
Put the file at `latex/figures/tls_2d_spectra_inhomogeneous.pdf`.

Figure 4.3: Sequence of 2D spectra of an inhomogeneously broadened qubit ensemble for increasing waiting time T. Compared to Fig. 4.2, static disorder produces an elongated diagonal feature, while homogeneous dynamics govern the antidiagonal width. (This placeholder should be replaced by the corresponding figure set.)

4.2 Dimer (Two Coupled Sites)

As the next step, a dimer is considered: two optically active sites with a possible coherent coupling constant J. This is the minimal model that can produce cross peaks and coupling-induced spectral structure in 2D spectra, and it is a natural bridge to larger ensembles.

4.2.1 Manifolds and Basis States

The standard excitation-manifold basis is used. The ground state is $|g\rangle$. The singly excited manifold is spanned by two localized excitations $|1\rangle$ and $|2\rangle$ ("site basis"), and the doubly excited state is $|f\rangle \equiv |12\rangle$. This matches the $|g\rangle, |n\rangle, |f\rangle$ scheme discussed in Ch. 3 with $n \in \{1, 2\}$.

4.2.2 Hamiltonian with Coupling J

In the single-excitation manifold, the Hamiltonian is taken as

$$\hat{H}_{\mathrm{S}}^{(\mathrm{dimer})} = \varepsilon_1 \, |1\rangle\langle 1| + \varepsilon_2 \, |2\rangle\langle 2| + J\left(|1\rangle\langle 2| + |2\rangle\langle 1|\right) + (\varepsilon_1 + \varepsilon_2) \, |f\rangle\langle f| \, . \tag{4.6}$$

Setting $J = 0$ recovers two uncoupled transitions. Finite J produces excitonic eigenstates and characteristic peak splittings/cross peaks in 2D spectra.

4.2.3 Exciton Basis (Useful Limiting Picture)

For interpretation of spectra it is often useful to diagonalize the single-excitation block. Introducing the site-energy difference $\Delta\varepsilon = \varepsilon_1 - \varepsilon_2$, the two single-exciton eigenenergies are

$$\varepsilon_\pm = \tfrac{1}{2}(\varepsilon_1 + \varepsilon_2) \pm \sqrt{\left(\tfrac{\Delta\varepsilon}{2}\right)^2 + J^2}. \tag{4.7}$$

The corresponding eigenstates $|\pm\rangle$ interpolate between localized excitations (for $|J| \ll |\Delta\varepsilon|$) and fully delocalized superpositions (for $|J| \gg |\Delta\varepsilon|$). In the 2D spectrum this coherent mixing is visible already at waiting time $T = 0$ through peak splittings and, generically, non-vanishing off-diagonal structure.

4.2.4 Dipole Operator

The dipole operator is written as the sum over allowed one-quantum transitions,

$$\hat{\mu}_{\mathrm{S}}^{(\mathrm{dimer})} = \sum_{n=1}^{2} \Big(\mu_{gn} \, |g\rangle\langle n| + \mu_{ng} \, |n\rangle\langle g| \Big) + \sum_{n=1}^{2} \Big(\mu_{nf} \, |n\rangle\langle f| + \mu_{fn} \, |f\rangle\langle n| \Big), \tag{4.8}$$

with $\mu_{ng} = \mu_{gn}^*$ and $\mu_{fn} = \mu_{nf}^*$. The $|n\rangle \leftrightarrow |f\rangle$ transitions are required to include excited-state absorption pathways in third-order response.

The specific dimer parameter choices and the figures to be shown for the dimer will be specified later.

4.2.5 Physical Interpretation in 2D Spectra: Coupling vs. Transfer

This dimer is the minimal setting in which the 2D spectrum can distinguish qualitatively different "excitation-transfer" mechanisms:

- **Coherent coupling (excitonic mixing).** For $J \neq 0$ the optical transitions are between $|g\rangle$ and the exciton states $|\pm\rangle$ (and, if included, between $|\pm\rangle$ and $|f\rangle$). In this case cross peaks can be present already at $T = 0$ because the eigenstates are delocalized and the Liouville pathways do not cancel exactly.

- **Incoherent population transfer.** If the sites are only weakly coupled (or even uncoupled) but the environment induces population transfer $|1\rangle \to |2\rangle$ (or both directions), then cross peaks are typically *absent at* $T = 0$ and *grow with* T as population migrates during the waiting time. In this scenario diagonal peak amplitudes decay while the corresponding cross peak rises on the same timescale.

This "$T = 0$ cross peak" versus "growth with T" distinction is a convenient rule of thumb when interpreting simulated (or measured) spectra.

4.2.6 Transfer Model During the Waiting Time

To model unidirectional or bidirectional excitation transfer within the single-excitation manifold during the waiting interval T, the coherent Hamiltonian evolution may be complemented by Lindblad jump operators of the form

$$\hat{L}_{2\leftarrow 1} = \sqrt{k_{2\leftarrow 1}} \, |2\rangle\langle 1| \,, \qquad \hat{L}_{1\leftarrow 2} = \sqrt{k_{1\leftarrow 2}} \, |1\rangle\langle 2| \,, \tag{4.9}$$

in addition to site-local relaxation/dephasing operators (Ch. 2). In the numerical workflow this affects the evolution superoperator acting between the second and third interactions and therefore controls the waiting-time dependence of diagonal and cross-peak amplitudes.

4.2.7 Coherences During T (Oscillatory Peak Dynamics)

If the system supports excited-state coherences during the waiting time (e.g. between $|+\rangle$ and $|-\rangle$), then selected peak amplitudes can acquire oscillatory T-dependence at the exciton splitting frequency (with a decay set by the corresponding dephasing rate). In practice, this provides a direct numerical handle to separate "population transfer" signatures (monotonic rise/decay) from coherence signatures (beats) in the same model.

4.3 Generalization to an N-Site Ensemble with Geometry

Finally, an ensemble of N optically active sites ("atoms" or chromophores) placed in a prescribed geometry is considered. Motivated by protofilament arrangements in microtubules, the geometry of interest in this thesis is a discretized cylindrical surface [10]. The model is formulated in the excitation-manifold basis $|g\rangle, \{|n\rangle\}, \{|f\rangle\}$ already used in Ch. 3.

4.3.1 States and Truncation

The ground state is $|g\rangle$. The single-excitation manifold is spanned by states $|n\rangle$, $n \in \{1, \ldots, N\}$, which represent one excitation localized at site n (or, after diagonalization, an excitonic eigenstate). If double excitations are included, states $|f\rangle \equiv |mn\rangle$ with $1 \leq m < n \leq N$ are used, representing two simultaneous excitations. In this thesis the Hilbert space is restricted to the ground state plus the single-excitation manifold, or (when needed for ESA contributions) to ground + single + double excitations.

4.3.2 What 2D Spectra Diagnose in Ensembles

In an N-site setting, 2D spectra are particularly useful because they separate several effects that are difficult to disentangle in linear absorption:

- **Static disorder (inhomogeneous broadening)** produces elongation along the diagonal in the $(\omega_{\text{coh}}, \omega_{\text{det}})$ plane, reflecting a distribution of transition frequencies across realizations.
- **Homogeneous dephasing** controls the antidiagonal width (for fixed disorder), i.e. he width perpendicular to the diagonal, and therefore provides access to dephasing timescales even in disordered ensembles.
- **Transport and transfer** redistribute amplitude between diagonal and off-diagonal features as a function of T. The pattern of emerging cross peaks encodes which states (or spatial regions) become correlated during the waiting time.

These qualitative diagnostics motivate the use of 2D-FTS observables as a readout for geometry-dependent excitation migration in the cylindrical architectures studied below.

4.3.3 Hamiltonian

In this basis, a generic Frenkel-exciton-type Hamiltonian reads

$$\hat{H}_{\text{S}}^{(N)} = \sum_{n=1}^{N} \varepsilon_n \, |n\rangle\langle n| + \sum_{m<n}^{N} J_{mn} \left(|m\rangle\langle n| + |n\rangle\langle m| \right) + \sum_{m<n}^{N} (\varepsilon_m + \varepsilon_n) \, |mn\rangle\langle mn| + \cdots , \tag{4.10}$$

where ε_n are site transition frequencies and J_{mn} are coherent couplings. The ellipsis indicates additional terms in the double-excitation manifold if one wishes to include interactions beyond an additive two-exciton energy (e.g. exciton–exciton shifts).

4.3.4 Geometry and Couplings

The couplings J_{mn} may be parameterized phenomenologically (e.g. nearest-neighbour) or computed from transition-dipole interactions. In the latter case one uses

$$J_{mn} \propto \frac{1}{\|\boldsymbol{r}_{mn}\|^3} \left[\boldsymbol{\mu}_m \cdot \boldsymbol{\mu}_n - 3 \, (\boldsymbol{\mu}_m \cdot \hat{\boldsymbol{r}}_{mn})(\boldsymbol{\mu}_n \cdot \hat{\boldsymbol{r}}_{mn}) \right] , \tag{4.11}$$

with site positions $\boldsymbol{r}_n$ and displacement $\boldsymbol{r}_{mn} = \boldsymbol{r}_m - \boldsymbol{r}_n$ [59–61].

For a cylindrical arrangement, let N_θ be the number of sites per ring and N_z the number of rings (so $N = N_\theta N_z$), with sites indexed by an azimuthal index and an axial index. Periodic boundary conditions can be imposed in the azimuthal direction.

4.3.5 Dipole Operator

The dipole operator is the sum over allowed one-quantum transitions between adjacent excitation manifolds,

$$\hat{\mu}_{\mathrm{S}}^{(N)} = \sum_{n=1}^{N} \Big(\mu_{gn} \, |g\rangle\langle n| + \mu_{ng} \, |n\rangle\langle g| \Big) + \sum_{m<n}^{N} \sum_{\ell \in \{m,n\}} \Big(\mu_{\ell,mn} \, |\ell\rangle\langle mn| + \mu_{mn,\ell} \, |mn\rangle\langle \ell| \Big), \tag{4.12}$$

where the second sum is only needed if the double-excitation manifold is included.

This model class captures collective transport pathways and their photon-echo signatures, offering a route to connect microscopic couplings and geometry to observable 2D spectra [10].

IDEA: Include this:

Multilevel Generalization

Realistic systems typically have many optically accessible excited states, and a central goal of this thesis is to extend the simulation framework from a minimal two-level model to multilevel Hamiltonians. A convenient (and widely used) idealisation that already captures the essential physics of ground-state bleach (GSB), stimulated emission (SE), and excited-state absorption (ESA) is a three-manifold "band" picture:

- a single ground state $|g\rangle$.
- a manifold of singly excited states $\{|n\rangle\}$.
- a manifold of higher excited (doubly excited) states $\{|f\rangle\}$.

Optically allowed transitions connect $|g\rangle \leftrightarrow |n\rangle$ and $|n\rangle \leftrightarrow |f\rangle$. In third-order spectroscopy the Liouville pathways do not contain populations of the f manifold (they end in coherences such as ρ_{fn}), so relaxation out of f typically enters effectively through additional dephasing/broadening of the $n \leftrightarrow f$ coherences.

Propagators in the secular approximation. In the secular approximation, the τ_2 evolution is expressed in terms of Liouville-space propagator elements of the generic form

$$U_{nn,mm}(\tau_2), \qquad U_{nm,nm}(\tau_2), \qquad U_{gg,nn}(\tau_2), \tag{4.13}$$

describing population transfer within the excited manifold ($nn \to mm$), coherence dephasing within the excited manifold ($nm \to nm$), and refilling of the ground state from excited populations ($nn \to gg$). ESA pathways additionally involve propagators for f-manifold coherences (e.g. $U_{fn,fm}(\tau_2)$).

Recipe for constructing multilevel signals. Given these propagators, multilevel spectra are obtained by the same "stitching" procedure as in the two-level case:

1. choose the relevant pulse sequence (and thus the allowed pathway classes and phase matching / phase cycling).
2. for each pathway, write the product of four dipole matrix elements (e.g. d_{ng}, d_{mg}, and, for ESA, d_{fn}) and three propagators (coherence–population–coherence) appropriate to that pathway.

3. Fourier transform over the detection time to obtain contributions as functions of ω_{det} (and, for 2D spectra, also over the coherence time to obtain ω_{coh}).

4. sum over all intermediate state indices $n, m, f, \ldots$.

This multilevel bookkeeping is exactly what becomes necessary once the system Hamiltonian contains multiple excited eigenstates (or excitonic bands) and the open-system Liouvillian generates non-trivial transfer and dephasing within these manifolds. The explicit multilevel formulation will therefore serve as the bridge from the pedagogical two-level picture to the full numerical simulations targeted in the later thesis chapters.

General dissipative channels (preview). For later use it is helpful to note that a multilevel relaxation model can be written compactly in Lindblad form by introducing one jump operator per allowed transition $|n\rangle \to |m\rangle$,

$$\hat{L}_{mn} = \sqrt{k_{mn}}\,|m\rangle\langle n|\,,$$

and, if needed, pure-dephasing operators for each level,

$$\hat{L}_n^{(\varphi)} = \sqrt{\gamma_{\varphi,n}}\,|n\rangle\langle n|\,.$$

These ingredients define the Liouvillian whose propagator elements $U_{ab,cd}(t)$ enter directly in the pathway products described above.

IMPORTANT: at the correct point implement this:

In a chosen basis $\{|n\rangle\}$ of the system Hilbert space Eq. (3.2) becomes

$$\boldsymbol{P}(t) = \sum_{n,m} \boldsymbol{\mu}_{nm}\,\rho_{S,mn}(t), \qquad \boldsymbol{\mu}_{nm} = \langle n|\,\hat{\boldsymbol{\mu}}\,|m\rangle\,.$$

Oscillating polarisation is generated by optical coherences, i.e. off-diagonal elements of the density matrix. If the relevant electronic states carry no permanent dipoles in the chosen eigenbasis, i.e. $\boldsymbol{\mu}_{nn} = \mathbf{0}$, then populations do not contribute.

Appendix A

Numerical Implementation

This chapter presents the computational framework developed for simulating one- and two-dimensional electronic spectroscopy of open quantum systems.

The implementation is built around the `qspectro2d` Python package, which provides a modular, configuration-driven workflow that bridges the theoretical framework established in previous chapters with practical computational methods for investigating quantum coherence phenomena in complex molecular systems. The open quantum system dynamics are solved using `QuTiP` [25].

A.1 Software Architecture and Design Philosophy

The `qspectro2d` package follows a modular design that separates concerns across distinct functional domains. The architecture enables efficiency through parallel processing and flexible configuration management. The core design is built around a **configuration-driven workflow** where YAML-based parameter specification decouples physical models from computational implementation. This approach ensures that **modular components** including atomic systems, laser pulses, environmental baths, and spectroscopic calculations remain independently configurable. The framework provides **scalable execution** with support for both local testing and high-performance computing (HPC) batch processing, complemented by **comprehensive post-processing** through automated workflows for averaging, Fourier transforms, and visualization.

The package structure comprises five main submodules: `core` (fundamental system components), `spectroscopy` (calculation engines), `config` (parameter management), `utils` (data handling utilities), and `visualization` (plotting tools).

A.2 Configuration-Driven Simulation Workflow

The simulation framework employs YAML configuration files to specify all physical and computational parameters, enabling reproducible simulations. A typical configuration defines the **atomic system** through the number of atoms, transition frequencies, geometric arrangement, and inhomogeneous broadening parameters. **Laser parameters** encompass pulse amplitudes, durations, carrier frequencies, and rotating wave approximation settings, while **environmental coupling** is characterized by bath type (Ohmic, Drude-Lorentz), temperature, coupling strength, and cutoff frequencies. Finally, **computational settings** specify the ODE solver choice (Lindblad vs. Bloch-Redfield), signal types, and time windows.

The workflow consists of three main execution phases: simulation (`calc_datas.py`), post-processing (`process_datas.py`), and visualization (`plot_datas.py`). This separation enables efficient parameter studies where computationally expensive simulations are performed once, followed by rapid exploration of different analysis and visualization options.

A.3 Third-Order Polarisation and Phase Cycling

In the numerical implementation, the electric field for multiple pulses is modeled following the formalism established in Ch. 3. The total electric field is expressed as

$$\boldsymbol{E}(\boldsymbol{r},t)=\sum_{m=1}^{3}\boldsymbol{E}_m(\boldsymbol{r},t)+\text{c.c.},$$

where c.c. denotes the complex conjugate, and

$$\boldsymbol{E}_m(\boldsymbol{r},t)=\boldsymbol{\chi}_m E'_m(t-\tau_m)\exp(-i(\omega_m t+\varphi_m)),$$

with frequency $\omega_m=2\pi\nu_m$, wavevector $\boldsymbol{k}_m$, and electric field amplitude $\boldsymbol{\chi}_m$. The envelope $E'_m(t-\tau_m)$, centered at τ_m, is assumed to be Gaussian with FWHM τ_p

$$E'_m(t-\tau_m)=\exp\left(-\frac{4\ln 2(t-\tau_m)^2}{\tau_p^2}\right).$$

For numerical efficiency, the Gaussian pulse is truncated to zero once the instantaneous intensity $|E'_m(t)|^2$ falls below 10^{-4} of its peak value (approximately two FWHM from the pulse center). In these field-free intervals, the evolution is propagated with the time-independent Liouvillian, which is computationally more efficient. The core spectroscopic calculation implements phase-cycled third-order polarisation measurement following the established four-wave mixing protocols detailed in Ch. 3, Sec. 3.6 [3, 54].

This thesis instead uses a more microscopic, non-perturbative (NP) approach: the full (time-dependent) equation of motion is solved with the pulses included and the induced polarisation is computed via Eq. (3.2). To isolate the nonlinear signal in the time domain, auxiliary runs are performed in which only a subset of pulses is active, and the lower-order (linear) contributions are subtracted from the full signal. In the weak-field regime and for centrosymmetric media (where $\chi^{(2)}$ vanishes), this yields a background-free polarisation that is dominated by the desired third-order response,

$$\boldsymbol{P}^{(3)}(t)\approx\boldsymbol{P}_{\text{total}}(t)-\sum_{m=1}^{3}\boldsymbol{P}_{(m)}(t), \tag{A.1}$$

where $\boldsymbol{P}_{\text{total}}(t)$ is obtained with all three pulses present and $\boldsymbol{P}_{(m)}(t)$ denotes the polarisation obtained when only pulse m is applied. In practice, the complex analytic (positive-frequency) component of the polarisation is used to retain amplitude and phase information for subsequent phase cycling and signal construction.

A.4 Open Quantum System Evolution

The quantum system evolution is handled by the `SimulationModuleOQS` class, which provides a unified interface for different solver backends implementing the open quantum system dynamics developed in Ch. 2. The implementation supports both Lindblad master equation evolution (via QuTiP's `mesolve`) and Bloch-Redfield dynamics (`brmesolve`) with automatic handling of pulse overlap and time-dependent Hamiltonians.

For time-dependent pulse sequences, the total Hamiltonian takes the form established in the spectroscopy theory (Ch. 3):

$$H(t)=H_0+\sum_i H_{\text{int},i}(t), \tag{A.2}$$

where H_0 is the bare system Hamiltonian and $H_{\text{int},i}(t)$ represents the interaction with pulse i, following the light-matter interaction formalism detailed in Ch. 3. The solver automatically detects pulse overlap regions and constructs appropriate piecewise evolution operators, ensuring accurate treatment of multi-pulse sequences regardless of timing complexity.

A.5 Inhomogeneous Broadening and Statistical Averaging

Real molecular systems exhibit inhomogeneous broadening due to environmental variations, as discussed in the spectroscopy fundamentals (Ch. 3). The implementation models this through Gaussian-distributed transition frequencies following the theoretical treatment:

$$p(\omega) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\omega-\omega_0)^2}{2\sigma^2}\right), \tag{A.3}$$

where $\sigma = \Delta_{\text{FWHM}}/(2\sqrt{2\ln 2})$ relates the standard deviation to the experimentally relevant full width at half maximum, consistent with the broadening models established in Ch. 3. The sampling employs a rejection algorithm optimized for computational efficiency while maintaining statistical accuracy.

A.6 Parallel Processing and Scalable Execution

The framework supports both local execution for testing and HPC cluster deployment for large parameter studies. The local workflow (`calc_datas.py`) generates all combinations of coherence times and inhomogeneous samples, while the HPC workflow (`hpc_batch_dispatch.py`) distributes individual parameter combinations across compute nodes.

For each combination of parameters, the averaged response is calculated following the inhomogeneous averaging procedure established in the spectroscopy theory (Ch. 3):

$$\langle E_{\text{S}}(t_{\text{coh}}, t_{\text{det}})\rangle = \frac{1}{N_{\text{inhom}}} \sum_{i=1}^{N_{\text{inhom}}} E_{\text{S}}^{(i)}(t_{\text{coh}}, t_{\text{det}}), \tag{A.4}$$

where N_{inhom} represents the number of inhomogeneous samples as defined in the broadening treatment (Ch. 3). The implementation utilizes Python's `ProcessPoolExecutor` for local parallel execution and integrates with SLURM-based job scheduling systems for HPC deployment, enabling efficient scaling from desktop testing to production-scale parameter sweeps.

A.7 Post-Processing and Spectral Analysis

The post-processing pipeline (`process_datas.py`) handles the conversion from time-domain signals to 2D spectra. **Data aggregation** groups individual parameter combination files by inhomogeneous sample and stacks them along coherence time dimensions. **Zero-padding** extends the time-domain data to improve frequency resolution according to:

$$E_{\text{extended}}(t) = \begin{cases} E(t) & \text{for } t \in [0, T_{\text{orig}}] \\ 0 & \text{for } t \in (T_{\text{orig}}, T_{\text{extended}}] \end{cases}. \tag{A.5}$$

Subsequently, **Fourier transformation** converts the data to frequency domain. For the rephasing signal this is implemented as the (two-dimensional) Fourier transform over the coherence and detection times, In practice, windowing trades spectral resolution for reduced ringing, while zero-padding improves visual interpolation without adding new information.

$$R_{\mathrm{R}}(\omega_1, T, \omega_3) \;=\; \int_0^{\infty} \mathrm{d}t_1 \int_0^{\infty} \mathrm{d}t_3 \; e^{+i\omega_1 t_1} \, e^{+i\omega_3 t_3} \, R_{\mathrm{R}}(t_1, T, t_3), \tag{A.6}$$

with the specific sign conventions matching those used in the spectroscopy chapter.

The frequency axes are expressed in wavenumber units (10^4 cm^{-1}) for direct comparison with experimental literature.

A.8 Model System Implementation

The `AtomicSystem` class provides flexible construction of multi-atom quantum systems with configurable geometries. For the microtubule-inspired models central to this thesis, the implementation supports **cylindrical arrangements** where atoms are positioned on cylindrical surfaces with specified chain and ring numbers. **Excitation manifold truncation** restricts the system to single and double excitation subspaces with automatic basis construction, while **dynamic coupling** enables nearest-neighbor and long-range interactions based on geometric proximity. The framework also incorporates **inhomogeneous disorder** through site-specific frequency variations sampled from realistic distributions.

The system construction automatically handles basis transformations between site and excitation manifold representations, enabling efficient evolution in the appropriate eigenbasis while maintaining physical interpretability of results.

IMPORTANT: implement this at a suitable location, of course only if it is not yet included:

Besides phase cycling, another common numerical strategy is to isolate a desired nonlinear order by combining runs with different subsets of pulses turned on (and subtracting lower-order contributions). In this thesis phase cycling is used, because it selects the desired rephasing/nonrephasing pathways directly through their phase dependence.

Appendix B

Dyson Series and Response Functions (Matter Side)

This appendix derives explicit, causal expressions for the n-th order Polarisation. It keeps the full tensorial formulation where the polarisation $\boldsymbol{P}^{(n)}(t)$ is a vector and the response $R^{(n)}_{\alpha\beta_1\cdots\beta_n}$ is a rank-(n+1) tensor. Starting from a driven master equation for the reduced density operator $\hat{\rho}_\mathrm{S}(t)$, the field dependence is isolated in the interaction picture and expanded in powers of the field. This leads to response functions, which capture the intrinsic material dynamics, and to convolution formulas for $P^{(n)}_\alpha$ that separate material properties from pulse shapes.

B.1 Driven Master Equation

The starting point is a driven open-system equation of motion for the reduced density operator $\hat{\rho}_\mathrm{S}(t)$ of the material system after tracing out the environment,

$$\frac{\mathrm{d}\hat{\rho}_\mathrm{S}(t)}{\mathrm{d}t} = -\mathrm{i}\,\mathcal{K}[\hat{\rho}_\mathrm{S}(t)] \quad + \mathrm{i}\,\mathcal{V}_\beta[\hat{\rho}_\mathrm{S}(t)]\,E_\beta(t), \tag{B.1}$$

where $\mathcal{K}$ is the field-free reduced generator obtained after eliminating the environment. It contains the Hamiltonian evolution of the system as well as dissipation and dephasing effects, as in Redfield or Lindblad theory (cf. Section 2.1). The driving field $\boldsymbol{E}(t)$ enters through its Cartesian components $E_\beta(t)$ and the corresponding dipole superoperators

$$\mathcal{V}_\beta[\hat{A}] \equiv [\hat{\mu}_\beta, \hat{A}], \tag{B.2}$$

with $\hat{\mu}_\beta$ the Cartesian components of the dipole operator $\hat{\boldsymbol{\mu}}$. Throughout this appendix repeated Cartesian indices are summed over (Einstein convention).

The commutator in Eq. (B.2) makes explicit that each interaction has a left and a right action on the density operator. This is the algebraic origin of Liouville-space pathways: whether $\hat{\mu}_\beta$ acts on the ket or on the bra determines which coherences or populations are created at each interaction. The commutator introduces a relative minus sign between the two actions.

The field-free propagator is defined as

$$\mathcal{U}_0(t) \equiv \exp(-\mathrm{i}\,\mathcal{K}t),$$

so that in the absence of the field the state evolves as

$$\hat{\rho}_\mathrm{S}(t) = \mathcal{U}_0(t - t_0)[\hat{\rho}_\mathrm{S}(t_0)].$$

B.2 Interaction Picture Formulation

To isolate the field-driven part of the evolution, introduce an interaction picture with respect to the field-free generator $\mathcal{K}$. Define the interaction-picture density operator by

$$\hat{\rho}_{\mathrm{S,I}}(t) \equiv \mathcal{U}_0\big(-(t-t_0)\big)\,[\hat{\rho}_{\mathrm{S}}(t)], \tag{B.3}$$

and the corresponding time-dependent dipole superoperators by

$$\mathcal{V}_{\mathrm{I},\beta}(t) \equiv \mathcal{U}_0\big(-(t-t_0)\big)\,\mathcal{V}_\beta\,\mathcal{U}_0(t-t_0). \tag{B.4}$$

By differentiating Eq. (B.3) with respect to time, inserting the master equation Eq. (B.1), and using that $\mathcal{U}_0$ commutes with $\mathcal{K}$, the equation of motion reduces to

$$\frac{\mathrm{d}\hat{\rho}_{\mathrm{S,I}}(t)}{\mathrm{d}t} = \mathrm{i}\,\mathcal{V}_{\mathrm{I},\beta}(t)[\hat{\rho}_{\mathrm{S,I}}(t)]\,E_\beta(t). \tag{B.5}$$

This form is exact for the reduced system. The field-free dynamics have been absorbed into the picture transformation and into the explicit time dependence of $\mathcal{V}_{\mathrm{I},\beta}(t)$. The only remaining term on the right-hand side is the light–matter coupling, which makes Eq. (B.5) a convenient starting point for perturbation theory in the field strength.

B.3 Dyson Series Expansion

Convert Eq. (B.5) into an integral equation by integrating from an initial time t_0 to t,

$$\hat{\rho}_{\mathrm{S,I}}(t) = \hat{\rho}_{\mathrm{S,I}}(t_0) \quad + \mathrm{i}\int_{t_0}^{t} \mathrm{d}\tau\,\mathcal{V}_{\mathrm{I},\beta}(\tau)[\hat{\rho}_{\mathrm{S,I}}(\tau)]\,E_\beta(\tau). \tag{B.6}$$

The initial condition is the same in both pictures, $\hat{\rho}_{\mathrm{S,I}}(t_0) = \hat{\rho}_{\mathrm{S}}(t_0)$.

To extract contributions of definite order in the field, substitute Eq. (B.6) back into itself iteratively. After n substitutions, collect all terms containing n factors of E_β. This produces the Dyson series (cf. Section 2.1.3), which is an infinite perturbation expansion in the field strength,

$$\hat{\rho}_{\mathrm{S,I}}(t) = \left[\mathbb{1} + \sum_{n=1}^{\infty}(\mathrm{i})^n \int_{t_0}^{t}\mathrm{d}\tau_1 \int_{t_0}^{\tau_1}\mathrm{d}\tau_2 \cdots \int_{t_0}^{\tau_{n-1}}\mathrm{d}\tau_n\,\mathcal{V}_{\mathrm{I},\beta_1}(\tau_1)\cdots\mathcal{V}_{\mathrm{I},\beta_n}(\tau_n)\,E_{\beta_1}(\tau_1)\cdots E_{\beta_n}(\tau_n)\right][\hat{\rho}_{\mathrm{S}}(t_0)]. \tag{B.7}$$

The nested limits $t_0 < \tau_n < \cdots < \tau_2 < \tau_1 < t$ enforce time ordering and therefore causality. This equation is useful for nonlinear spectroscopy because it isolates the n-th order term that can be selected experimentally.

B.4 Polarisation as a Perturbation Series

The macroscopic polarisation is defined as the expectation value of the dipole operator (cf. Equation (3.2)),

$$P_\alpha(t) = \mathrm{Tr}\{\hat{\mu}_\alpha\,\hat{\rho}_{\mathrm{S}}(t)\}\,. \tag{B.8}$$

Using the interaction-picture representation of the state and cyclicity of the trace, it can be written as

$$P_\alpha(t) = \mathrm{Tr}\{\hat{\mu}_{\mathrm{I},\alpha}(t)\,\hat{\rho}_{\mathrm{S,I}}(t)\} = \sum_{n=0}^{\infty} P_\alpha^{(n)}(t),$$

where $\hat{\mu}_{\mathrm{I},\alpha}(t)$ is the dipole operator evolved under the field-free dynamics. In Liouville-space notation this is the adjoint action of the field-free propagator on the observable,

$$\hat{\mu}_{\mathrm{I},\alpha}(t) = \mathcal{U}_0^\dagger(t-t_0)[\hat{\mu}_\alpha].$$

The n-th order contribution is

$$P_\alpha^{(n)}(t) = \mathrm{Tr}\Big\{\hat{\mu}_{\mathrm{I},\alpha}(t)\,\hat{\rho}_{\mathrm{S,I}}^{(n)}(t)\Big\},$$

with $\hat{\rho}_{\mathrm{S,I}}^{(n)}$ the term containing n factors of the field.

Substituting the Dyson expansion Eq. (B.7) and extracting the n-th order term yields an expression that is naturally written using the field-free propagator $\mathcal{U}_0$ and the Schrödinger-picture superoperators $\mathcal{V}_\beta$. Repeatedly insert the definition Eq. (B.4) into the ordered product $\mathcal{V}_{\mathrm{I},\beta_1}(\tau_1)\cdots\mathcal{V}_{\mathrm{I},\beta_n}(\tau_n)$, use the group property $\mathcal{U}_0(a)\mathcal{U}_0(b) = \mathcal{U}_0(a+b)$ to combine adjacent propagators, and move the remaining picture transformations within the trace using cyclicity. This produces free-propagation intervals between successive interaction events and the final interval between the last interaction and the measurement time. The result is

$$\begin{aligned}
P_\alpha^{(n)}(t) = (\mathrm{i})^n \int_{t_0}^{t} \mathrm{d}\tau^{(1)} \cdots \int_{t_0}^{\tau^{(n-1)}} \mathrm{d}\tau^{(n)} \\
\times \mathrm{Tr}\Big\{\hat{\mu}_\alpha\,\mathcal{U}_0(t-\tau^{(1)})\,\mathcal{V}_{\beta_1}\,\mathcal{U}_0(\tau^{(1)}-\tau^{(2)}) \\
\mathcal{V}_{\beta_2}\cdots\mathcal{V}_{\beta_n}\,\mathcal{U}_0(\tau^{(n)}-t_0)[\hat{\rho}_{\mathrm{S}}(t_0)]\Big\} \\
\times \prod_{j=1}^{n} E_{\beta_j}(\tau^{(j)}).
\end{aligned} \tag{B.9}$$

Reading from right to left inside the trace, the system starts in the initial state $\hat{\rho}_{\mathrm{S}}(t_0)$, undergoes n alternating interactions (via $\mathcal{V}_{\beta_j}$) and field-free evolutions (via $\mathcal{U}_0$), and is finally probed through the dipole operator $\hat{\mu}_\alpha$. This sequence underlies the usual Liouville-space pathway picture.

B.5 Definition of the Response Function

The expression Eq. (B.9) is written in terms of absolute interaction times $\tau^{(j)}$. For spectroscopy it is often more convenient to use nonnegative delay intervals between successive events. Define

$$\tau_n = t - \tau^{(1)}, \qquad \tau_{n-j} = \tau^{(j)} - \tau^{(j+1)}\ (j = 1,\dots,n-1),$$

so that each $\tau_j \geq 0$ is an interval between two adjacent interactions, and τ_n is the interval between the last interaction and the measurement at time t. Inverting these relations gives

$$\tau^{(n)} = t - \sum_{k=1}^{n}\tau_k, \qquad \tau^{(n-1)} = t - \sum_{k=2}^{n}\tau_k, \qquad \dots, \qquad \tau^{(1)} = t - \tau_n. \tag{B.10}$$

The Jacobian of the transformation is unity. In the limit $t_0 \to -\infty$, the nested integration region over absolute times maps to independent integrals over $\tau_j \in [0,\infty)$. This change of variables is conceptually depicted in Fig. B.1.

Substitute Eq. (B.10) into Eq. (B.9) and assume the system starts in equilibrium, $\hat{\rho}_{\mathrm{S}}(t_0) = \hat{\rho}_{\mathrm{eq}}$, at $t_0 \to -\infty$. The initial evolution is then trivial because $\mathcal{U}_0(t)[\hat{\rho}_{\mathrm{eq}}] = \hat{\rho}_{\mathrm{eq}}$. The n-th order polarisation becomes

$$\begin{aligned}
P_\alpha^{(n)}(t) = \int_0^\infty \mathrm{d}\tau_n \cdots \int_0^\infty \mathrm{d}\tau_1\, R^{(n)}_{\alpha\beta_1\cdots\beta_n}(\tau_n,\dots,\tau_1) \\
\times E_{\beta_1}(t-\tau_n)\,E_{\beta_2}(t-\tau_n-\tau_{n-1})\cdots E_{\beta_n}\left(t-\sum_{k=1}^{n}\tau_k\right).
\end{aligned} \tag{B.11}$$

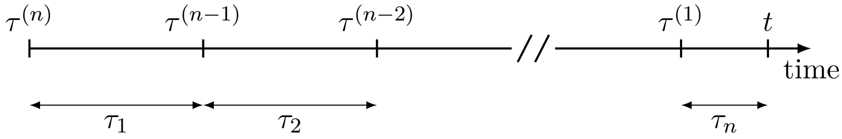

Figure B.1: Time variables in this appendix. **Old** (top): $\tau^{(j)}$ are absolute interaction times. **New** (bottom): τ_j are nonnegative delay intervals between successive events.

This formula separates the material dynamics from the experimental pulse shapes. The response tensor $R^{(n)}_{\alpha\beta_1\cdots\beta_n}$ is the field-independent factor,

$$R^{(n)}_{\alpha\beta_1\cdots\beta_n}(\tau_n,\ldots,\tau_1) \equiv (\mathrm{i})^n\,\mathrm{Tr}\Big\{\hat{\mu}_\alpha\,\mathcal{U}_0(\tau_n)\,\mathcal{V}_{\beta_1}\,\mathcal{U}_0(\tau_{n-1})\,\mathcal{V}_{\beta_2}\cdots\mathcal{V}_{\beta_n}\,\mathcal{U}_0(\tau_1)\,\mathcal{V}_{\beta_n}[\hat{\rho}_{\mathrm{eq}}]\Big\}. \tag{B.12}$$

By construction, $R^{(n)}_{\alpha\beta_1\cdots\beta_n}(\tau_n,\ldots,\tau_1)$ describes the intrinsic response of the system to n successive light–matter interactions separated by nonnegative delay intervals. Causality is explicit: the response vanishes whenever any $\tau_j < 0$, and Eq. (B.11) only involves field values at times earlier than t.

B.6 Fixed-Polarisation Projection Used in This Thesis

The derivation above is fully polarisation-resolved. In the numerical simulations reported in this thesis, all pulses are taken linearly polarised along a fixed laboratory direction $\boldsymbol{e}_{\mathrm{in}}$, and the detected signal is the component along $\boldsymbol{e}_{\mathrm{out}}$ (typically $\boldsymbol{e}_{\mathrm{out}} = \boldsymbol{e}_{\mathrm{in}}$). Because the simulations use a collinear setup with one fixed polarisation axis for excitation and readout, this projection is used throughout. Accordingly,

$$E_\beta(t) = e_{\mathrm{in},\beta}\,E(t), \qquad P(t) = e_{\mathrm{out},\alpha}\,P_\alpha(t), \tag{B.13}$$

so that the tensor response contracts to the scalar quantity

$$R^{(n)} \equiv e_{\mathrm{out},\alpha}\,R^{(n)}_{\alpha\beta_1\cdots\beta_n}\,e_{\mathrm{in},\beta_1}\cdots e_{\mathrm{in},\beta_n}. \tag{B.14}$$

In this fixed-polarisation setting, the equations used in the main text are

$$P^{(n)}(t) = \int_0^\infty \mathrm{d}\tau_n\cdots\int_0^\infty \mathrm{d}\tau_1\,R^{(n)}(\tau_n,\ldots,\tau_1)\,E(t-\tau_n)\,E(t-\tau_n-\tau_{n-1})\cdots E\Big(t-\sum_{k=1}^{n}\tau_k\Big), \tag{B.15}$$

with

$$R^{(n)}(\tau_n,\ldots,\tau_1) \equiv (\mathrm{i})^n\,\mathrm{Tr}\Big\{\hat{\mu}\,\mathcal{U}_0(\tau_n)\,\mathcal{V}\,\mathcal{U}_0(\tau_{n-1})\,\mathcal{V}\cdots\mathcal{V}\,\mathcal{U}_0(\tau_1)\,\mathcal{V}[\hat{\rho}_{\mathrm{eq}}]\Big\}. \tag{B.16}$$

Whenever response functions are used in the main text, indices are suppressed and $P^{(n)}$ and $R^{(n)}$ refer to these fixed-polarisation scalar projections. This is consistent with the numerical workflow, which uses a collinear setup (see Sec. 3.6) and therefore a fixed laboratory polarisation orientation.

Appendix C

Maxwell-to-Signal Derivation (Field Side)

This appendix gives the *field-side* derivation for coherent 2D-FTS/2DES. Starting from Maxwell's equations, it shows how the radiated, heterodyne-detected signal field is linear in the third-order polarisation $\boldsymbol{P}^{(3)}$. This provides the proportionality used in Sec. 3.3.

The argument proceeds in three steps. First, Maxwell's equations lead to a driven wave equation in which the macroscopic polarisation acts as the source. Second, under the slowly varying envelope approximation, the emitted signal field follows the corresponding component of $\boldsymbol{P}^{(3)}$. The microscopic expression for $\boldsymbol{P}^{(3)}$ is derived on the matter side in App. B.

C.1 From Maxwell's Equations to a Wave Equation for the Radiation Field

This section makes explicit why the macroscopic polarisation is the source of the emitted optical field [62]. The central point is that only the *transverse* part of the electromagnetic field propagates as radiation, and it is sourced by the transverse current. In condensed-matter optics, that current is naturally written in terms of the polarisation.

C.1.1 Maxwell's Equations and Electromagnetic Potentials

In natural units ($\hbar = c = 1$) and Heaviside–Lorentz form, Maxwell's equations in the presence of charge and current densities $\varrho(\boldsymbol{r}, t)$ and $\boldsymbol{j}(\boldsymbol{r}, t)$ read [**guidry1991gaugefieldtheories**]

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) = \boldsymbol{j}(\boldsymbol{r}, t) + \frac{\partial \boldsymbol{E}(\boldsymbol{r}, t)}{\partial t}, \tag{C.1}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}, t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r}, t)}{\partial t}, \tag{C.2}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}, t) = \varrho(\boldsymbol{r}, t), \tag{C.3}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}, t) = 0. \tag{C.4}$$

Here $\boldsymbol{E}$ is the electric field and $\boldsymbol{B}$ the magnetic field.

It is convenient to express $\boldsymbol{E}$ and $\boldsymbol{B}$ in terms of a vector potential $\boldsymbol{A}(\boldsymbol{r}, t)$ and a scalar potential $\phi(\boldsymbol{r}, t)$,

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \qquad \boldsymbol{E} = -\frac{\partial \boldsymbol{A}}{\partial t} - \boldsymbol{\nabla}\phi. \tag{C.5}$$

This automatically satisfies Faraday's law Eq. (C.2) and Gauss' law for magnetism Eq. (C.4), reducing the number of independent equations.

C.1.2 Gauge Freedom and Coulomb Gauge

To proceed, a gauge must be chosen. The potentials in Eq. (C.5) are not unique: for any scalar function $\chi(\boldsymbol{r}, t)$,

$$\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\chi, \qquad \phi' = \phi - \frac{\partial \chi}{\partial t}, \tag{C.6}$$

leaves $\boldsymbol{E}$ and $\boldsymbol{B}$ invariant. For optics, the Coulomb gauge is used,

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0.$$

Inserting Eq. (C.5) into Gauss' law Eq. (C.3) then gives the Poisson equation

$$\boldsymbol{\nabla}^2 \phi(\boldsymbol{r}, t) = -\varrho(\boldsymbol{r}, t), \tag{C.7}$$

so ϕ is determined instantaneously in space. This does not violate causality; it reflects the gauge-dependent description of the Coulomb interaction. The propagating optical field resides in the transverse sector, discussed next.

C.1.3 Transverse/Longitudinal Decomposition

To separate propagating radiation from the Coulomb part, decompose fields into transverse and longitudinal components. Only the transverse component carries radiative degrees of freedom.

Any sufficiently well-behaved vector field $\boldsymbol{F}(\boldsymbol{r})$ admits the Helmholtz decomposition,

$$\boldsymbol{F} = \boldsymbol{F}^{\perp} + \boldsymbol{F}^{\parallel}, \tag{C.8}$$

where the transverse part is divergence-free and the longitudinal part is curl-free,

$$\boldsymbol{\nabla} \cdot \boldsymbol{F}^{\perp} = 0, \qquad \boldsymbol{\nabla} \times \boldsymbol{F}^{\parallel} = 0. \tag{C.9}$$

Applying this decomposition to the electric field in Eq. (C.5),

$$\boldsymbol{E} = \boldsymbol{E}^{\perp} + \boldsymbol{E}^{\parallel}, \tag{C.10}$$

with $\boldsymbol{E}^{\perp} = -\partial_t \boldsymbol{A}$ in Coulomb gauge and $\boldsymbol{E}^{\parallel} = -\boldsymbol{\nabla}\phi$. Maxwell's equations Eqs. (C.1) to (C.4) then split into independent transverse and longitudinal parts,

$$\boldsymbol{\nabla} \times \boldsymbol{E}^{\perp} = -\frac{\partial \boldsymbol{B}}{\partial t}, \qquad \boldsymbol{\nabla} \times \boldsymbol{E}^{\parallel} = 0, \tag{C.11}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}^{\parallel} = \varrho, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{E}^{\perp} = 0. \tag{C.12}$$

Faraday's law constrains only the transverse field because $\boldsymbol{\nabla} \times \boldsymbol{E}^{\parallel} = 0$ identically, while Gauss' law fixes the longitudinal field through the charge density. The propagating and Coulomb degrees of freedom are therefore separated.

To understand which current sources what, recall that charge and current are not independent: they are linked by the continuity equation

$$\frac{\partial \varrho(\boldsymbol{r}, t)}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{j}(\boldsymbol{r}, t) = 0. \tag{C.13}$$

Decomposing $\boldsymbol{j} = \boldsymbol{j}^{\perp} + \boldsymbol{j}^{\parallel}$ and using $\boldsymbol{\nabla} \cdot \boldsymbol{j}^{\perp} = 0$ shows that Eq. (C.13) constrains only the longitudinal part,

$$\frac{\partial \varrho}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{j}^{\parallel} = 0. \tag{C.14}$$

In this sense, $\boldsymbol{j}^{\parallel}$ is tied to Coulomb near-field physics, while radiative fields are sourced by $\boldsymbol{j}^{\perp}$.

Two immediate consequences follow.

- Because $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$, the magnetic field is purely transverse, so $\boldsymbol{B} = \boldsymbol{B}^{\perp}$.
- In Coulomb gauge $\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0$, the vector potential is purely transverse, so $\boldsymbol{A} = \boldsymbol{A}^{\perp}$.

Combining the decomposition with Coulomb gauge and Eq. (C.5) yields

$$\boldsymbol{E}^{\perp} = -\frac{\partial \boldsymbol{A}}{\partial t}, \qquad \boldsymbol{E}^{\parallel} = -\boldsymbol{\nabla}\phi. \tag{C.15}$$

The radiation field relevant for propagating optics is $\boldsymbol{E}^{\perp}$. The longitudinal field $\boldsymbol{E}^{\parallel}$ is fixed instantaneously by ϱ via the Poisson equation and does not describe propagation.

C.1.4 Wave Equation for the Transverse Electric Field

With the radiative field isolated, the wave equation it satisfies is derived. Starting from Faraday's law Eq. (C.2), take the curl of both sides,

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{B}). \tag{C.16}$$

Next, eliminate $\boldsymbol{\nabla} \times \boldsymbol{B}$ using Ampère–Maxwell's law Eq. (C.1) to obtain

$$-\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = \frac{\partial^2 \boldsymbol{E}}{\partial t^2} + \frac{\partial \boldsymbol{j}}{\partial t}. \tag{C.17}$$

Only transverse parts contribute to the curl–curl operator because $\boldsymbol{\nabla} \times \boldsymbol{E}^{\parallel} = 0$. Likewise, the radiated field is sourced by the transverse current $\boldsymbol{j}^{\perp}$. Projecting onto the transverse sector gives

$$-\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}^{\perp}) = \frac{\partial^2 \boldsymbol{E}^{\perp}}{\partial t^2} + \frac{\partial \boldsymbol{j}^{\perp}}{\partial t}. \tag{C.18}$$

Using $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}^{\perp}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{E}^{\perp}) - \boldsymbol{\nabla}^2 \boldsymbol{E}^{\perp}$ and $\boldsymbol{\nabla} \cdot \boldsymbol{E}^{\perp} = 0$ yields the driven wave equation

$$\boldsymbol{\nabla}^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t) - \frac{\partial^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t)}{\partial t^2} = \frac{\partial \boldsymbol{j}^{\perp}(\boldsymbol{r}, t)}{\partial t}. \tag{C.19}$$

C.1.5 From Transverse Current to Polarisation

The wave equation Eq. (C.19) is expressed in terms of the current $\boldsymbol{j}^{\perp}$. In optics it is more convenient to work with the macroscopic polarisation $\boldsymbol{P}(\boldsymbol{r}, t)$, which describes the motion of bound charges in a neutral dielectric on coarse-grained scales. The associated current equals the time derivative of the polarisation,

$$\boldsymbol{j}(\boldsymbol{r}, t) = \frac{\partial \boldsymbol{P}(\boldsymbol{r}, t)}{\partial t}. \tag{C.20}$$

Together with the continuity equation Eq. (C.13), this implies the standard relation between bound charge density and polarisation,

$$\varrho_b(\boldsymbol{r}, t) = -\boldsymbol{\nabla} \cdot \boldsymbol{P}(\boldsymbol{r}, t). \tag{C.21}$$

Substituting the polarisation current Eq. (C.20) into Eq. (C.19) gives

$$\boldsymbol{\nabla}^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t) - \frac{\partial^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t)}{\partial t^2} = \frac{\partial^2 \boldsymbol{P}(\boldsymbol{r}, t)}{\partial t^2}. \tag{C.22}$$

Eq. (C.22) shows directly that the macroscopic polarisation is the source of the radiated transverse field. Measuring the emitted optical field therefore probes the polarisation induced by the incident light.

For nonlinear spectroscopy, write $\boldsymbol{P} = \boldsymbol{P}^{(1)} + \boldsymbol{P}_{\mathrm{NL}}$, where $\boldsymbol{P}_{\mathrm{NL}}$ collects higher-order terms such as $\boldsymbol{P}^{(3)}$. The linear response $\boldsymbol{P}^{(1)}$ is absorbed into an effective refractive index $n(\omega)$ with $n^2 = 1 + \chi^{(1)}$. With this, Eq. (C.22) becomes

$$\boldsymbol{\nabla}^2 \boldsymbol{E}(\boldsymbol{r}, t) - n^2 \frac{\partial^2}{\partial t^2} \boldsymbol{E}(\boldsymbol{r}, t) = \frac{\partial^2}{\partial t^2} \boldsymbol{P}_{\mathrm{NL}}(\boldsymbol{r}, t). \tag{C.23}$$

In a three-pulse experiment operated in the weak-field regime, the dominant nonlinear source is the third-order term,

$$\boldsymbol{P}_{\mathrm{NL}}(\boldsymbol{r}, t) \approx \boldsymbol{P}^{(3)}(\boldsymbol{r}, t), \tag{C.24}$$

where $\boldsymbol{P}^{(3)}$ is cubic in the incident field.

The remaining task is to connect $\boldsymbol{P}^{(3)}$ to a measurable signal field using the SVEA.

C.2 Slowly Varying Envelope Approximation and the Emitted Field

The driven wave equation Eq. (C.23) is solved to connect the nonlinear polarisation to the observed signal field. The slowly varying envelope approximation exploits the separation between the fast optical oscillation at carrier frequency ω_L and the slow evolution of envelopes. In this limit, the phase-matched signal field is proportional to the corresponding component of the nonlinear polarisation.

C.2.1 Setup: Fourier Components and Ansatz

Consider a slab sample of thickness L along z. The third-order polarisation contains Fourier components with different wavevector combinations $\boldsymbol{k}_{\mathrm{S}}$. Phase matching selects which signal direction is observed, as discussed in Sec. 3.5.2. A single component is written with a slowly varying envelope,

$$\boldsymbol{P}_{\mathrm{NL}}(\boldsymbol{r}, t) \supset \boldsymbol{P}_{\mathrm{S}}^{(3)}(t)\, e^{\mathrm{i}(\boldsymbol{k}_{\mathrm{S}} \cdot \boldsymbol{r} - \omega_L t)} + \mathrm{c.c.} \tag{C.25}$$

where $\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)$ varies slowly in time. A corresponding forward-propagating field solution is sought with matching carrier phase,

$$\boldsymbol{E}_{\mathrm{S}}(\boldsymbol{r}, t) = \tfrac{1}{2}\, \boldsymbol{E}_{\mathrm{S}}(z, t)\, e^{\mathrm{i}(k'_{\mathrm{S}} z - \omega_L t)} + \mathrm{c.c.}, \tag{C.26}$$

with dispersion relation $k'_{\mathrm{S}} = n(\omega_L)\omega_L$.

C.2.2 SVEA Conditions

The SVEA assumes that the envelopes $\boldsymbol{E}_{\mathrm{S}}(z, t)$ and $\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)$ vary slowly compared to the optical oscillation,

$$|\partial_t \boldsymbol{E}_{\mathrm{S}}| \ll \omega_L\, |\boldsymbol{E}_{\mathrm{S}}|, \qquad \left|\partial_z^2 \boldsymbol{E}_{\mathrm{S}}\right| \ll k'_{\mathrm{S}}\, |\partial_z \boldsymbol{E}_{\mathrm{S}}|. \tag{C.27}$$

This scale separation allows higher-order derivatives of the envelope to be neglected when they appear alongside rapidly oscillating carrier factors.

Spatial and Temporal Derivatives

To isolate the envelope dynamics, keep only the positive-frequency part and suppress explicit time dependence momentarily. For a positive-frequency component

$$\boldsymbol{E}_{\mathrm{S}}^{(+)}(z) = \tfrac{1}{2}\, \boldsymbol{E}_{\mathrm{S}}(z)\, e^{\mathrm{i}k'_{\mathrm{S}} z}, \tag{C.28}$$

one finds

$$\partial_z \boldsymbol{E}_{\mathrm{S}}^{(+)} = \tfrac{1}{2}\left(\partial_z \boldsymbol{E}_{\mathrm{S}} + \mathrm{i}k'_{\mathrm{S}}\boldsymbol{E}_{\mathrm{S}}\right)e^{\mathrm{i}k'_{\mathrm{S}}z}, \tag{C.29}$$

$$\partial_z^2 \boldsymbol{E}_{\mathrm{S}}^{(+)} = \tfrac{1}{2}\left(\partial_z^2 \boldsymbol{E}_{\mathrm{S}} + 2\mathrm{i}k'_{\mathrm{S}}\partial_z\boldsymbol{E}_{\mathrm{S}} - k'^2_{\mathrm{S}}\boldsymbol{E}_{\mathrm{S}}\right)e^{\mathrm{i}k'_{\mathrm{S}}z} \approx \tfrac{1}{2}\left(2\mathrm{i}k'_{\mathrm{S}}\partial_z\boldsymbol{E}_{\mathrm{S}} - k'^2_{\mathrm{S}}\boldsymbol{E}_{\mathrm{S}}\right)e^{\mathrm{i}k'_{\mathrm{S}}z}, \tag{C.30}$$

where the last step uses Eq. (C.27) to neglect $\partial_z^2\boldsymbol{E}_{\mathrm{S}}$.

For the time dependence, the analogous approximation is

$$\partial_t^2\left[\boldsymbol{E}_{\mathrm{S}}(z,t)e^{-\mathrm{i}\omega_L t}\right] \approx -\omega_L^2\,\boldsymbol{E}_{\mathrm{S}}(z,t)e^{-\mathrm{i}\omega_L t}, \qquad \partial_t^2\left[\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)e^{-\mathrm{i}\omega_L t}\right] \approx -\omega_L^2\,\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)e^{-\mathrm{i}\omega_L t}. \tag{C.31}$$

The carrier oscillation at ω_L dominates, so second derivatives of the envelopes can be dropped.

C.2.3 Envelope Propagation Equation

Insert Eq. (C.25) and Eq. (C.26) into Eq. (C.23). Using the SVEA derivative forms from Eq. (C.30) and Eq. (C.31), and imposing the dispersion relation $k'^2_{\mathrm{S}} = n^2(\omega_L)\omega_L^2$, the leading carrier terms cancel. The result is a first-order propagation equation for the envelope,

$$\mathrm{i}k'_{\mathrm{S}}\,\frac{\partial\boldsymbol{E}_{\mathrm{S}}(z,t)}{\partial z} = -\omega_L^2\,\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)\,e^{\mathrm{i}\Delta k\,z}, \tag{C.32}$$

where $\Delta k = k_{\mathrm{S}} - k'_{\mathrm{S}}$ is the phase mismatch between the polarisation wavevector and the propagation constant $k'_{\mathrm{S}} = n(\omega_L)\omega_L$. This equation shows how the field builds up through the sample, driven locally by the induced polarisation.

C.2.4 Integration and Sinc Factor

Equation (C.32) is integrated across the sample. With no signal at the entrance, $\boldsymbol{E}_{\mathrm{S}}(0,t) = 0$, integration from $z = 0$ to $z = L$ gives the exiting signal field,

$$\boldsymbol{E}_{\mathrm{S}}(L,t) = \mathrm{i}\,\frac{\omega_L}{n(\omega_L)}\,\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)\,L\,e^{\mathrm{i}\Delta kL/2}\,\mathrm{sinc}\left(\frac{\Delta kL}{2}\right). \tag{C.33}$$

This result shows that the emitted signal field is directly proportional to the corresponding component of $\boldsymbol{P}_{\mathrm{S}}^{(3)}$, and the sinc term captures phase-mismatch suppression. The response is maximal near $\Delta k \approx 0$.

When the experiment is phase matched, $\Delta k \approx 0$, so $\mathrm{sinc}(\Delta kL/2) \to 1$ and $e^{\mathrm{i}\Delta kL/2} \to 1$. Equation (C.33) then reduces to

$$\boldsymbol{E}_{\mathrm{S}}(t) \equiv \boldsymbol{E}_{\mathrm{S}}(L,t) \propto \mathrm{i}\,\frac{\omega_L}{n(\omega_L)}\,\boldsymbol{P}_{\mathrm{S}}^{(3)}(t)\,L. \tag{C.34}$$

In the frequency domain, this relation is often summarized as

$$\boldsymbol{E}_{\mathrm{S}}(\omega) = \kappa\,\mathrm{i}\omega\,\boldsymbol{P}^{(3)}(\omega), \tag{C.35}$$

where κ is a real prefactor that depends on propagation conventions and experimental geometry. Equation (C.35) is the Maxwell-side link used throughout: it makes the measurement a linear functional of $\boldsymbol{P}^{(3)}$.

This completes the field-side derivation. The incident fields drive the matter system, producing $\boldsymbol{P}^{(3)}$ via the response-function formalism in App. B. That polarisation then acts as the source term that radiates the measured signal field. Detection models and the construction of 2D spectra from phase-selected time-domain signals are discussed in Sec. 3.7.

Appendix D

Ultrashort-Pulse Intuition

This appendix collects an optional geometric intuition for the role of pulse carrier phases in pathway selection. It complements the formal treatment in Secs. 3.5.2 and 3.5.3 and Sec. 3.6 without changing the main derivation used in the thesis.

D.1 Carrier Phase, Pulse Area, and Bloch-Sphere Picture

The carrier phases ϕ_j in Eq. (3.25) are experimentally controlled parameters that imprint well-defined phase factors on driven coherences and hence on the nonlinear polarisation. This provides the pathway selection used throughout this thesis. This geometric picture is used as intuition. All quantitative results in this thesis are obtained from the full, numerically exact propagation and the phase-selection procedures developed in Sec. 3.6.

A visual interpretation arises for a resonantly driven two-level system within the RWA, where ϕ_j sets the rotation axis of the drive. Writing the complex Rabi frequency of pulse j as

$$\Omega_j(t) \equiv \boldsymbol{\mu}_{eg} \cdot \boldsymbol{E}_j(t), \tag{D.1}$$

the interaction Hamiltonian in the laser-rotating frame can be expressed as

$$\hat{H}_{\mathrm{int},j}(t) \;\propto\; \Omega_j(t-t_j)\Big(e^{-\mathrm{i}\phi_j}\;|e\rangle\langle g| + e^{+\mathrm{i}\phi_j}\;|g\rangle\langle e|\Big). \tag{D.2}$$

Changing ϕ_j rotates the complex drive while leaving the pulse envelope unchanged. On the Bloch sphere this corresponds to a rotation about an equatorial axis with azimuth angle ϕ_j (cf. [52]).

The pulse area

$$\theta_j \;\equiv\; \int_{-\infty}^{\infty} \mathrm{d}t\,\Omega_j(t-t_j) \tag{D.3}$$

sets the corresponding rotation angle. In the idealised delta-pulse limit one replaces $\Omega_j(t-t_j) \to \theta_j\,\delta(t-t_j)$, so the pulse acts as an instantaneous unitary

$$\hat{U}_j \;=\; \exp\!\left[-\frac{\mathrm{i}\,\theta_j}{2}\,\big(\cos\phi_j\,\hat{\tau}_x + \sin\phi_j\,\hat{\tau}_y\big)\right], \tag{D.4}$$

where $\hat{\tau}_{x,y,z}$ denote the Pauli matrices.

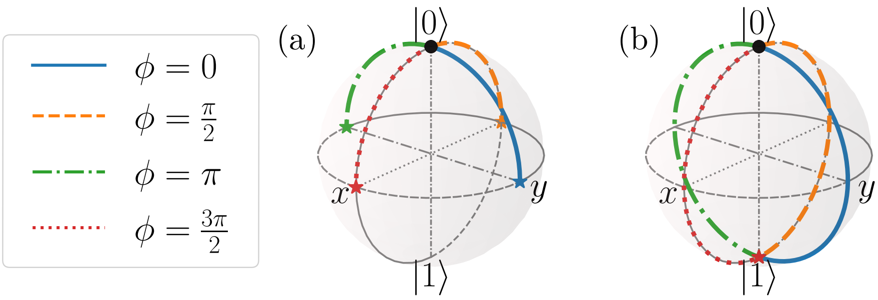

Figure D.1: Bloch-sphere visualization of one ultrashort interaction with a two-level system. The initial state is $|0\rangle$. Panels show examples for (a) $\theta = \pi/2$ and (b) $\theta = \pi$ with different carrier phase kicks ϕ.

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